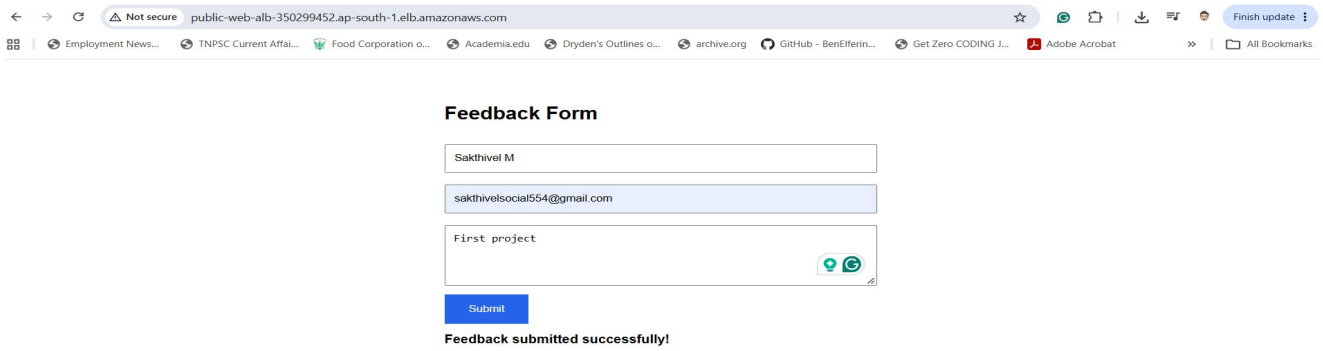
Project-01

Building Scalable, Secure, and High-Performance Web Applications with AWS 3-Tier Architecture

Item	Details
Name	SAKTHIVEL M
AWS region	Asia Pacific (Mumbai) ap-south-1
Application	Flask feedback app (Nginx web tier, Flask API tier, Amazon RDS MySQL)
Repository	github.com/Vennilavanguvi/Three-Tier-Architecture
Public URL (ALB DNS)	http://public-web-alb-350299452.ap-south-1.elb.amazonaws.com



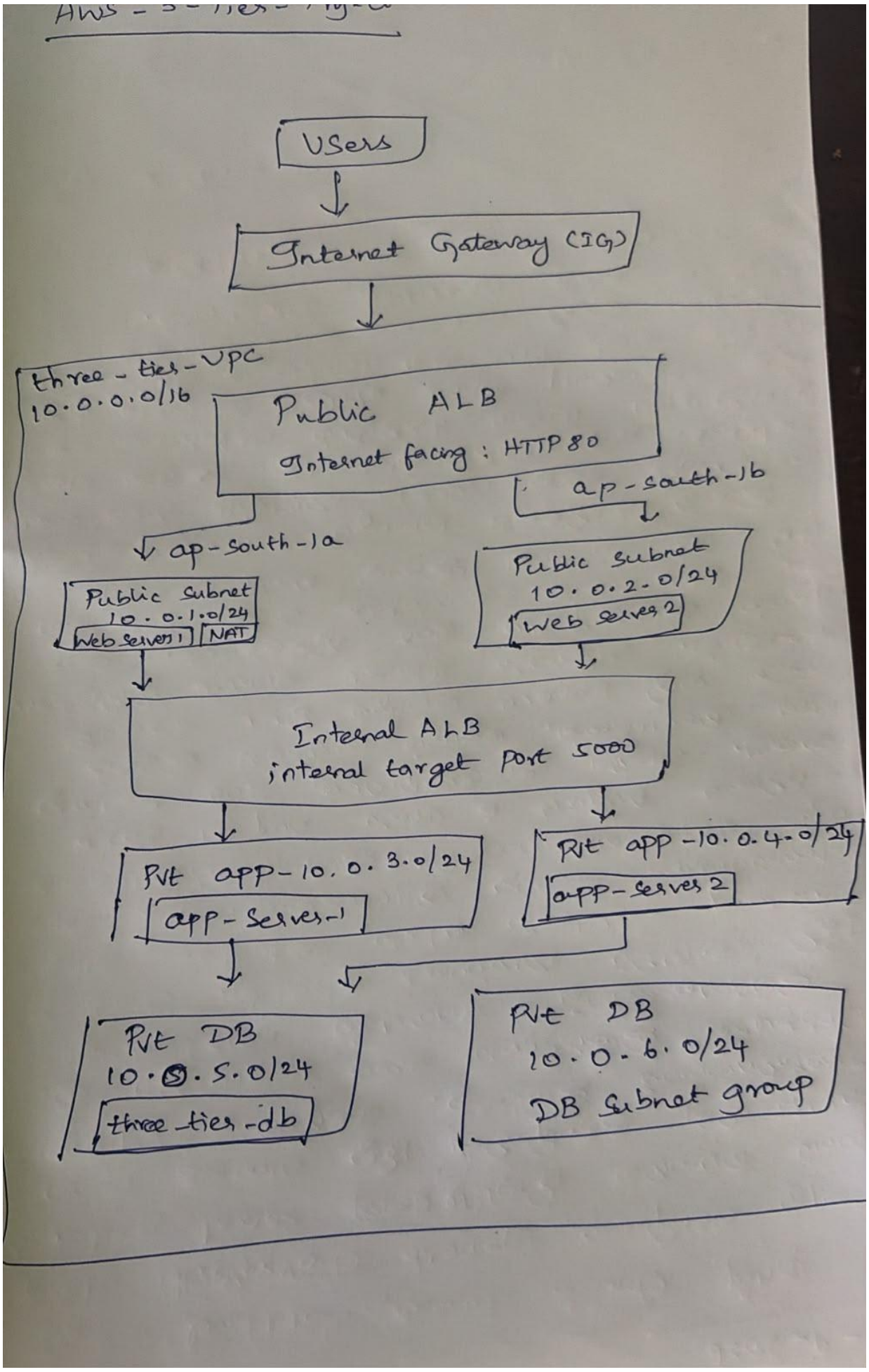
1. Project overview and architecture

The objective was to deploy a scalable, highly available and secure Python application on AWS using a 3-tier architecture. The web tier (Nginx) and the application tier (Flask API) each run on two EC2 instances in two Availability Zones behind load balancers, and the data tier is an Amazon RDS MySQL database in isolated private subnets.

Flow

Users → Internet gateway → public ALB → Nginx web servers → internal ALB → Flask app servers (port 5000) → Amazon RDS MySQL (port 3306)

AWS - 3 - Tier - Arch



Resources

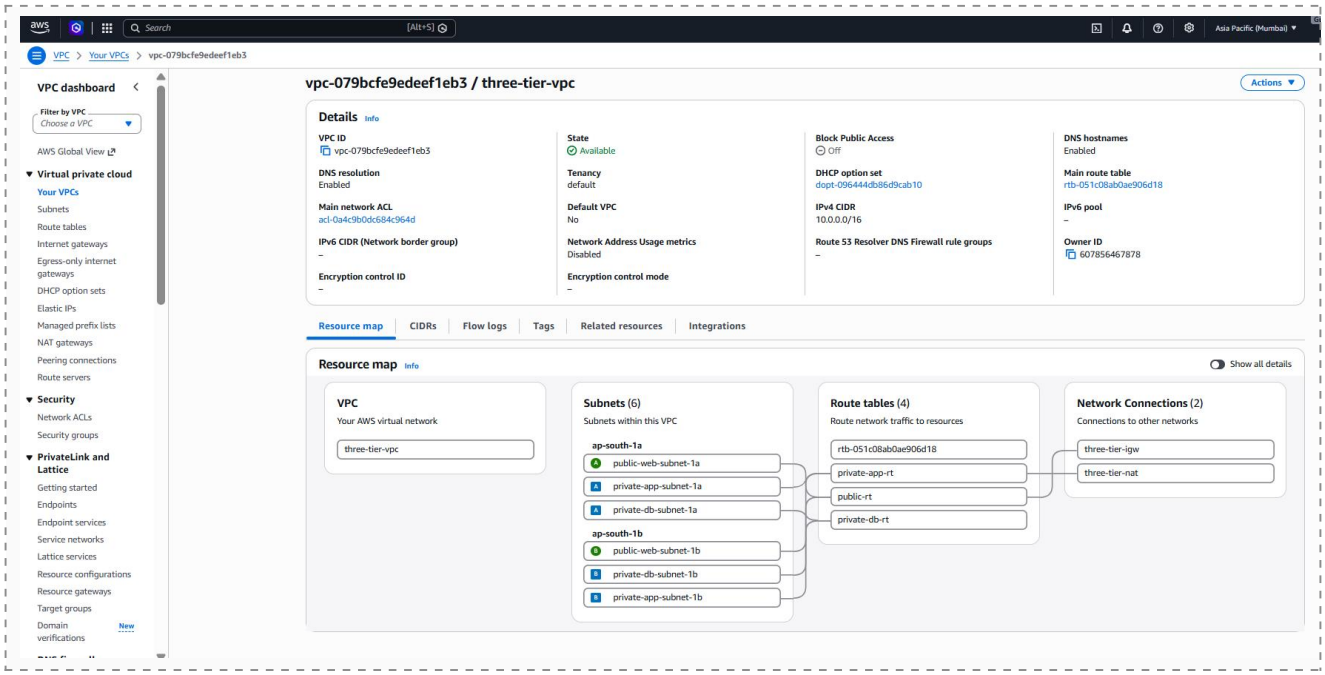
Resource	Name	ID / value
VPC	three-tier-vpc	vpc-079bcfe9edeef1eb3 (10.0.0.0/16)
Internet gateway	three-tier-igw	igw-0cef97bdee8b07336
Elastic IP	three-tier-nat-eip	13.203.0.161 (eipalloc-03ec95abe6202bb9d)
NAT gateway	three-tier-nat	nat-04a05ec3cf4c780d3 (zonal, public-web-subnet-1a)
Public ALB	public-web-alb	public-web-alb-350299452.ap-south-1.elb.amazonaws.com
Internal ALB	app-internal-alb	internal-app-internal-alb-1148195198.ap-south-1.elb.amazonaws.com
Target group (web)	web-tg	HTTP 80, health check /
Target group (app)	app-tg	HTTP 5000, health check /
RDS instance	three-tier-db	three-tier-db.cbkmimim2279.ap-south-1.rds.amazonaws.com
RDS engine	MySQL Community 8.4.11	db.t4g.micro, 20 GiB, single-AZ, not public
Web servers	web-server-1, web-server-2	Ubuntu 24.04, t3.micro, public subnets 1a / 1b
App servers	app-server-1, app-server-2	app-server-1: i-09f14fbaab5b9320c, app-server-2: i-0bba524a100ad278d
IAM role	ec2-ssm-role	AmazonSSMManagedInstanceCore (Session Manager access)
AMIs	web-server-ami, app-server-ami	Used to clone the second server in each tier

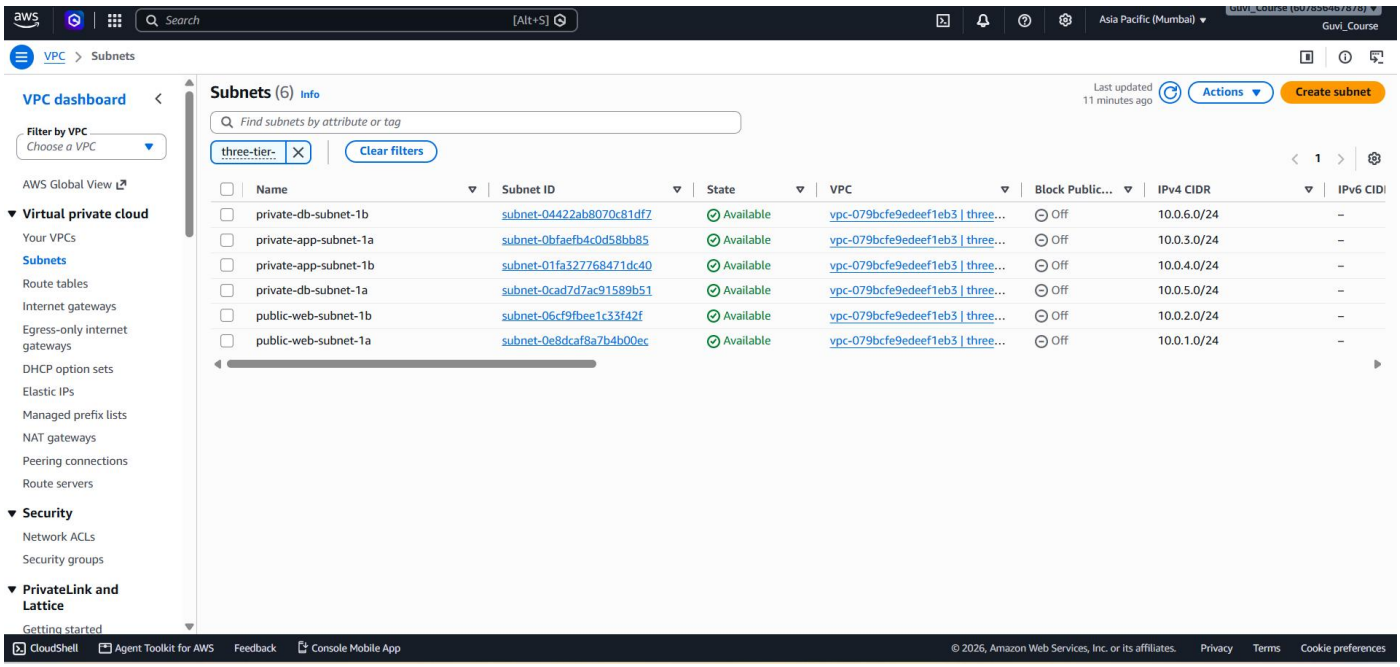
2. Task 1: AWS networking infrastructure

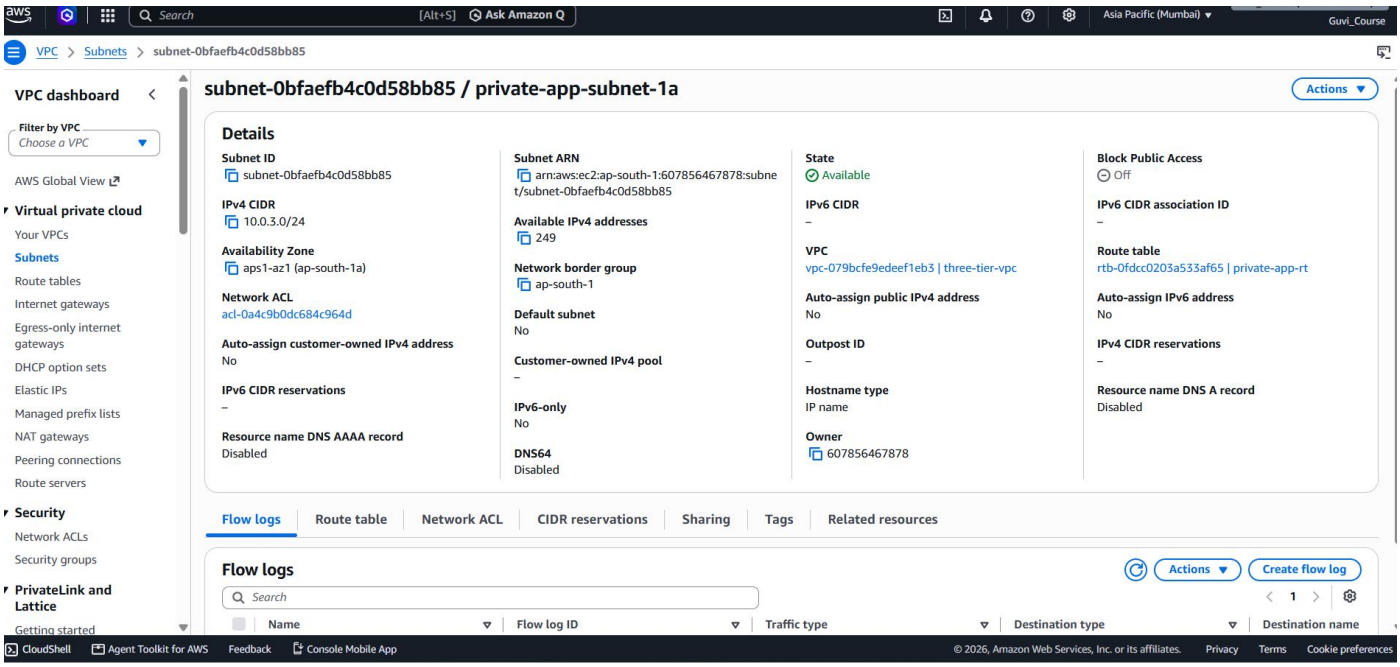
All networking resources were created manually in the VPC console. The VPC uses a /16 range and /24 subnets, so each subnet provides 256 addresses.

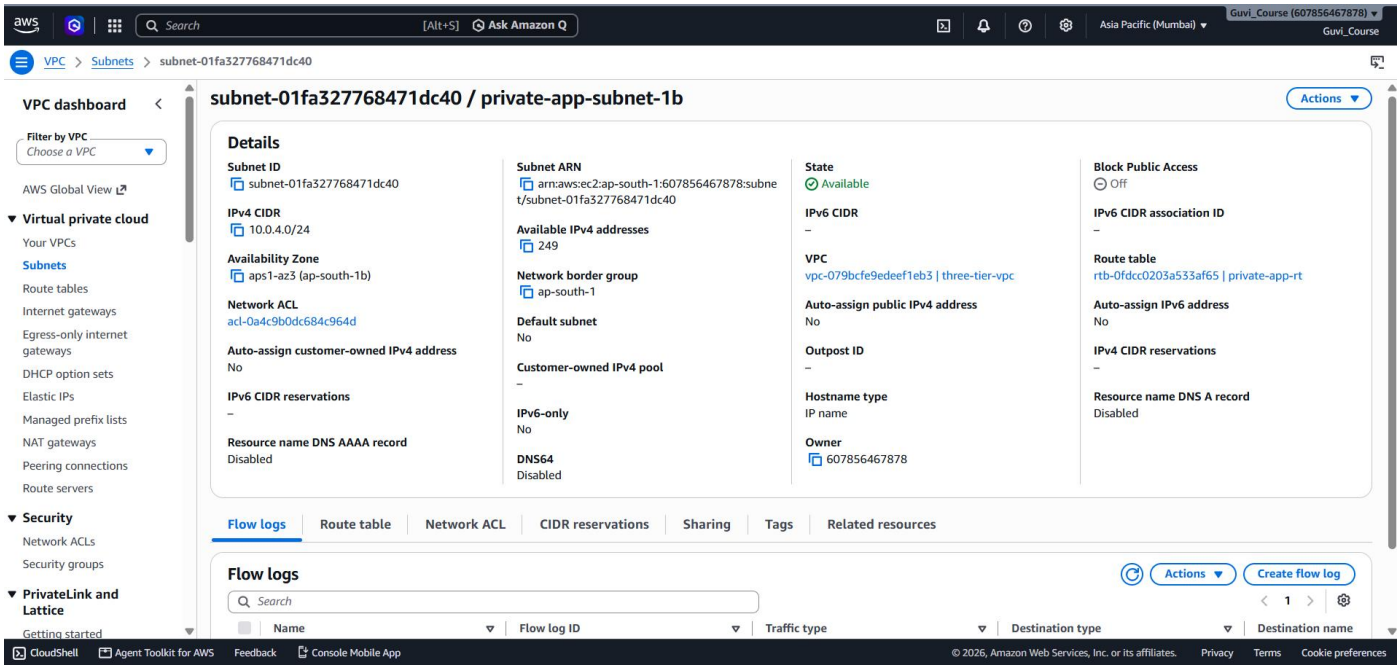
VPC and subnets

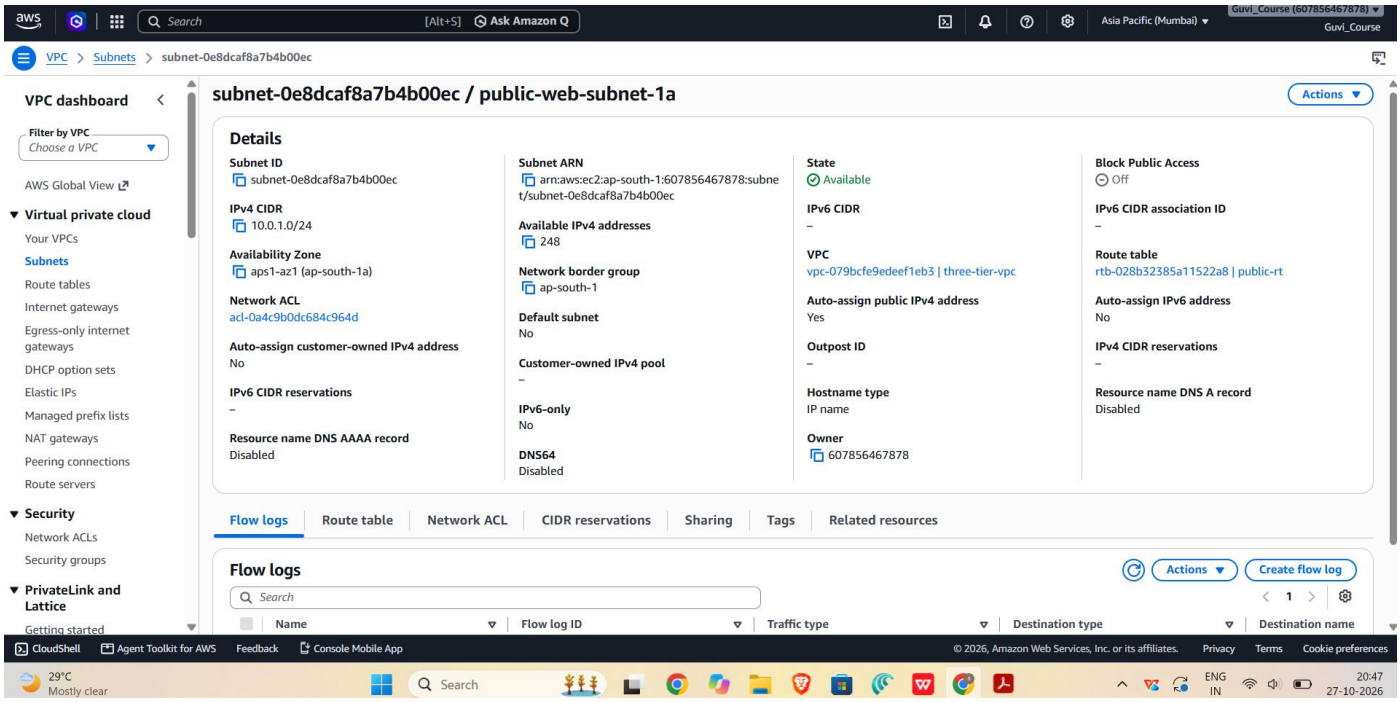
Subnet name	Tier	AZ	CIDR	Subnet ID
public-web-subnet-1a	Public (web)	ap-south-1a	10.0.1.0/24	`subnet-0e8dcaf8a7b4b00ec`
public-web-subnet-1b	Public (web)	ap-south-1b	10.0.2.0/24	`subnet-06cf9fbee1c33f42f`
private-app-subnet-1a	Private (app)	ap-south-1a	10.0.3.0/24	`subnet-0bfaefb4c0d58bb85`
private-app-subnet-1b	Private (app)	ap-south-1b	10.0.4.0/24	`subnet-01fa327768471dc40`
private-db-subnet-1a	Private (database)	ap-south-1a	10.0.5.0/24	`subnet-0cad7d7ac91589b51`
private-db-subnet-1b	Private (database)	ap-south-1b	10.0.6.0/24	`subnet-04422ab8070c81df7`

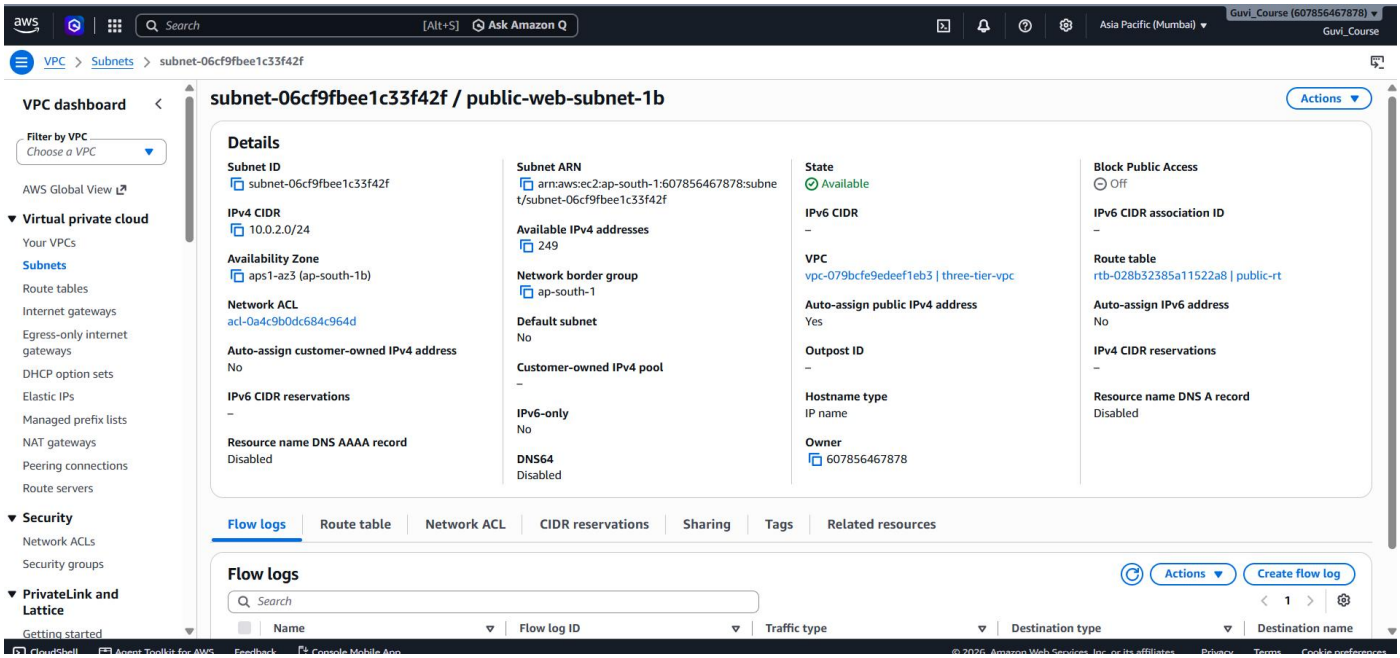


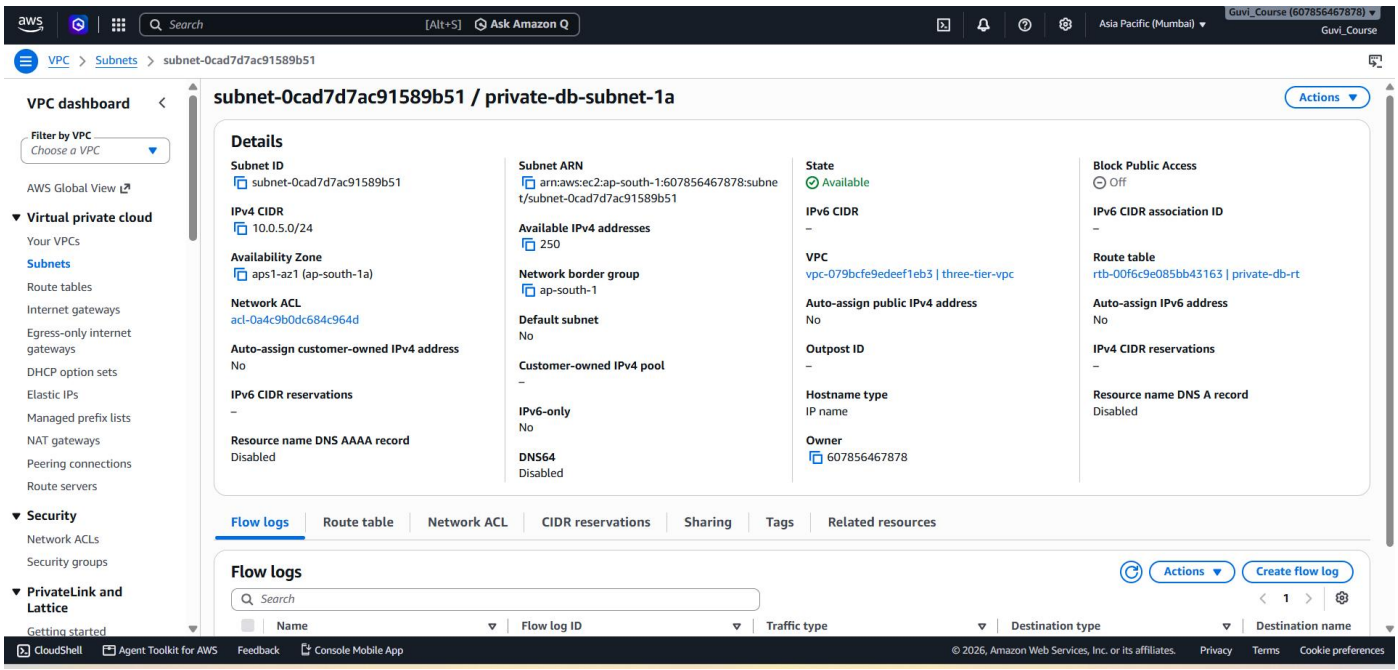

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IPv4 CIDR
IPv6 CIDI
private-db-subnet-1b subnet-04422ab8070c81df7 Available vpc-079bcfe9edeef1eb3 | three... Off 10.0.6.0/24
private-app-subnet-1a subnet-0bfaefb4c0d58bb85 Available vpc-079bcfe9edeef1eb3 | three... Off 10.0.3.0/24
private-app-subnet-1b subnet-01fa327768471dc40 Available vpc-079bcfe9edeef1eb3 | three... Off 10.0.4.0/24
private-db-subnet-1a subnet-0cad7d7ac91589b51 Available vpc-079bcfe9edeef1eb3 | three... Off 10.0.5.0/24
public-web-subnet-1b subnet-06cf9fbee1c33f42f Available vpc-079bcfe9edeef1eb3 | three... Off 10.0.2.0/24
public-web-subnet-1a subnet-0e8dcaf8a7b4b00ec Available vpc-079bcfe9edeef1eb3 | three... Off 10.0.1.0/24
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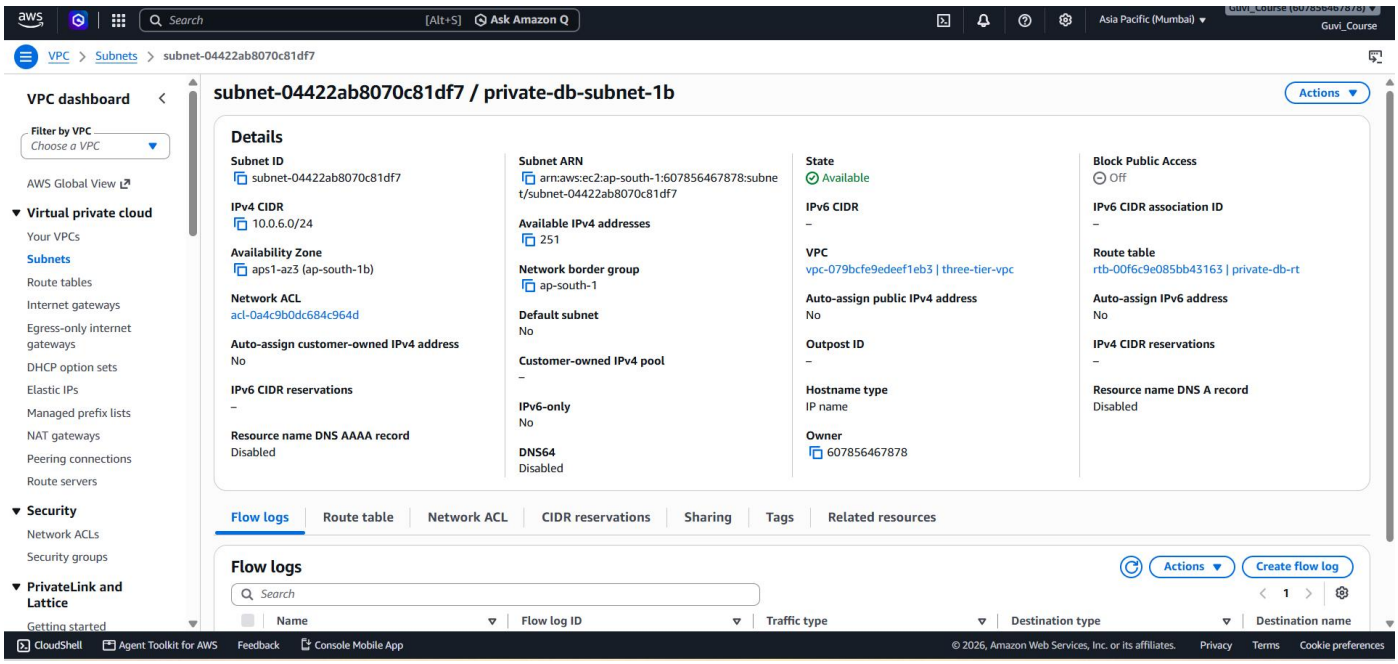

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subnet-0bfaefb4c0d58bb85
IPv4 CIDR
10.0.3.0/24
Availability Zone
aps1-az1 (ap-south-1a)
Network ACL
acl-0a4c9b0dc684c964d
Auto-assign customer-owned IPv4 address
No
IPv6 CIDR reservations
–
Resource name DNS AAAA record
Disabled
Subnet ARN
arn:aws:ec2:ap-south-1:607856467878:subnet/subnet-0bfaefb4c0d58bb85
Available IPv4 addresses
249
Network border group
ap-south-1
Default subnet
No
Customer-owned IPv4 pool
–
IPv6-only
No
DNS64
Disabled
State
Available
IPv6 CIDR
–
VPC
vpc-079bcfe9edeef1eb3 | three-tier-vpc
Auto-assign public IPv4 address
No
Outpost ID
–
Hostname type
IP name
Owner
607856467878
Block Public Access
Off
IPv6 CIDR association ID
–
Route table
rtb-0fdcc0203a533af65 | private-app-rt
Auto-assign IPv6 address
No
IPv4 CIDR reservations
–
Resource name DNS A record
Disabled
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subnet-01fa327768471dc40
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10.0.4.0/24
Availability Zone
aps1-az3 (ap-south-1b)
Network ACL
acl-0a4c9b0dc684c964d
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arn:aws:ec2:ap-south-1:607856467878:subnet/subnet-01fa327768471dc40
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VPC
vpc-079bcfe9edeef1eb3 | three-tier-vpc
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Outpost ID
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Hostname type
IP name
Owner
607856467878
Block Public Access
Off
IPv6 CIDR association ID
–
Route table
rtb-0fdcc0203a533af65 | private-app-rt
Auto-assign IPv6 address
No
IPv4 CIDR reservations
–
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Disabled
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Flow log ID
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Destination type
Destination name
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Available IPv4 addresses
248
Network border group
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Default subnet
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IPv6-only
No
DNS64
Disabled
State
Available
IPv6 CIDR
–
VPC
vpc-079bcfe9edeef1eb3 | three-tier-vpc
Auto-assign public IPv4 address
Yes
Outpost ID
–
Hostname type
IP name
Owner
607856467878
Block Public Access
Off
IPv6 CIDR association ID
–
Route table
rtb-028b32385a11522a8 | public-rt
Auto-assign IPv6 address
No
IPv4 CIDR reservations
–
Resource name DNS A record
Disabled
Flow logs
Route table
Network ACL
CIDR reservations
Sharing
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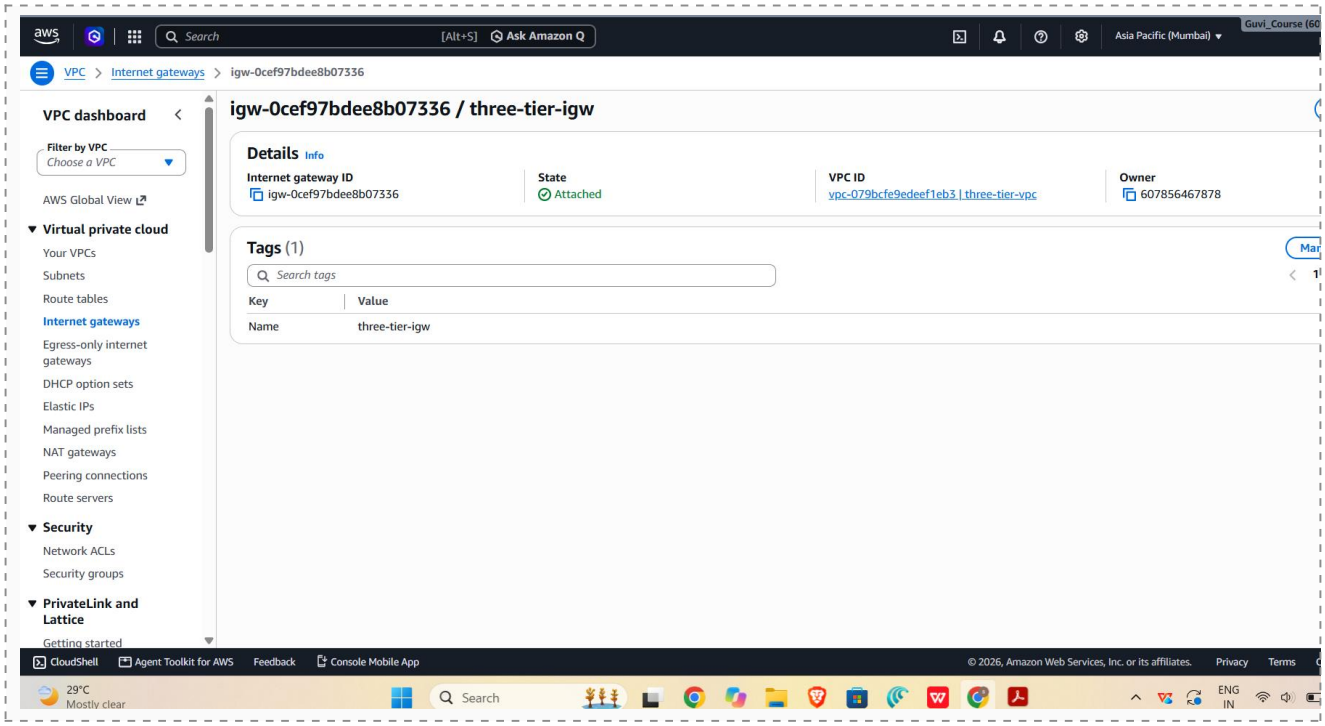

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IPv6-only
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vpc-079bcfe9edeef1eb3 | three-tier-vpc
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Hostname type
IP name
Owner
607856467878
Block Public Access
Off
IPv6 CIDR association ID
–
Route table
rtb-028b32385a11522a8 | public-rt
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Traffic type
Destination type
Destination name

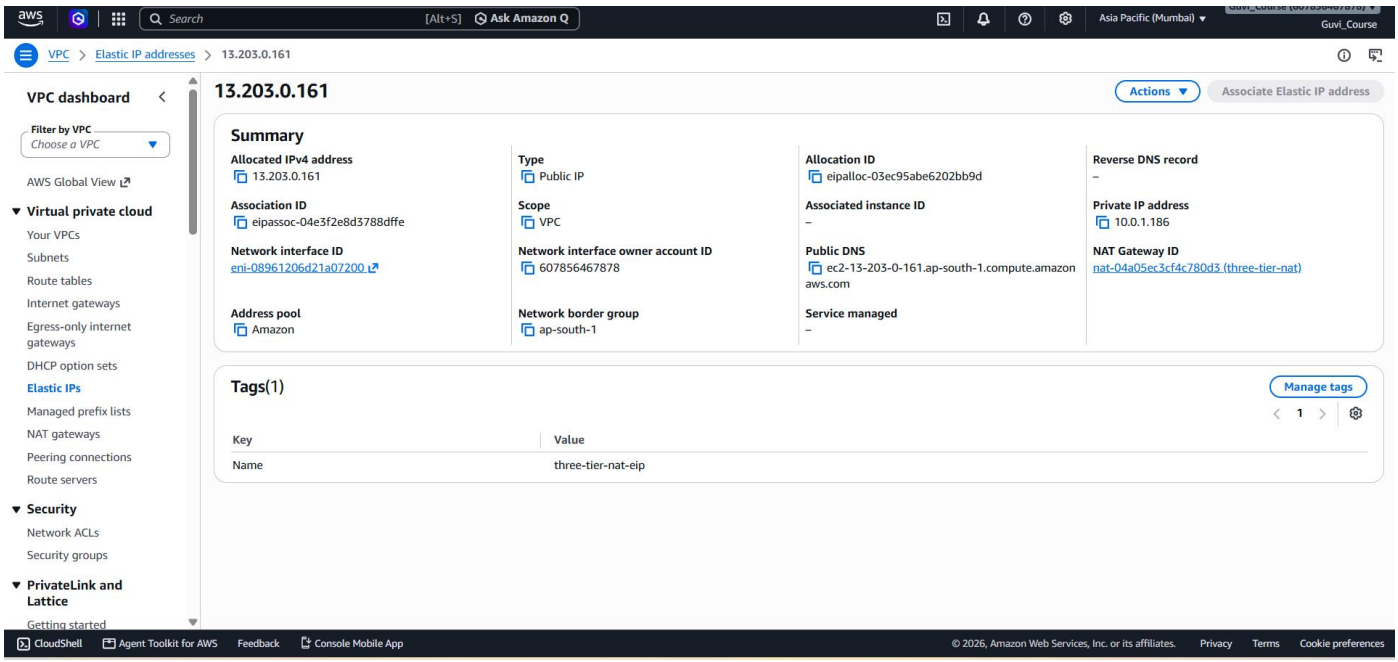

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Availability Zone
aps1-az1 (ap-south-1a)
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IPv6 CIDR reservations
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Available IPv4 addresses
250
Network border group
ap-south-1
Default subnet
No
Customer-owned IPv4 pool
–
IPv6-only
No
DNS64
Disabled
State
Available
IPv6 CIDR
–
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Outpost ID
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Hostname type
IP name
Owner
607856467878
Block Public Access
Off
IPv6 CIDR association ID
–
Route table
rtb-00f6c9e085bb43163 | private-db-rt
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No
IPv4 CIDR reservations
–
Resource name DNS A record
Disabled
Flow logs
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Internet gateway, Elastic IP and NAT gateway

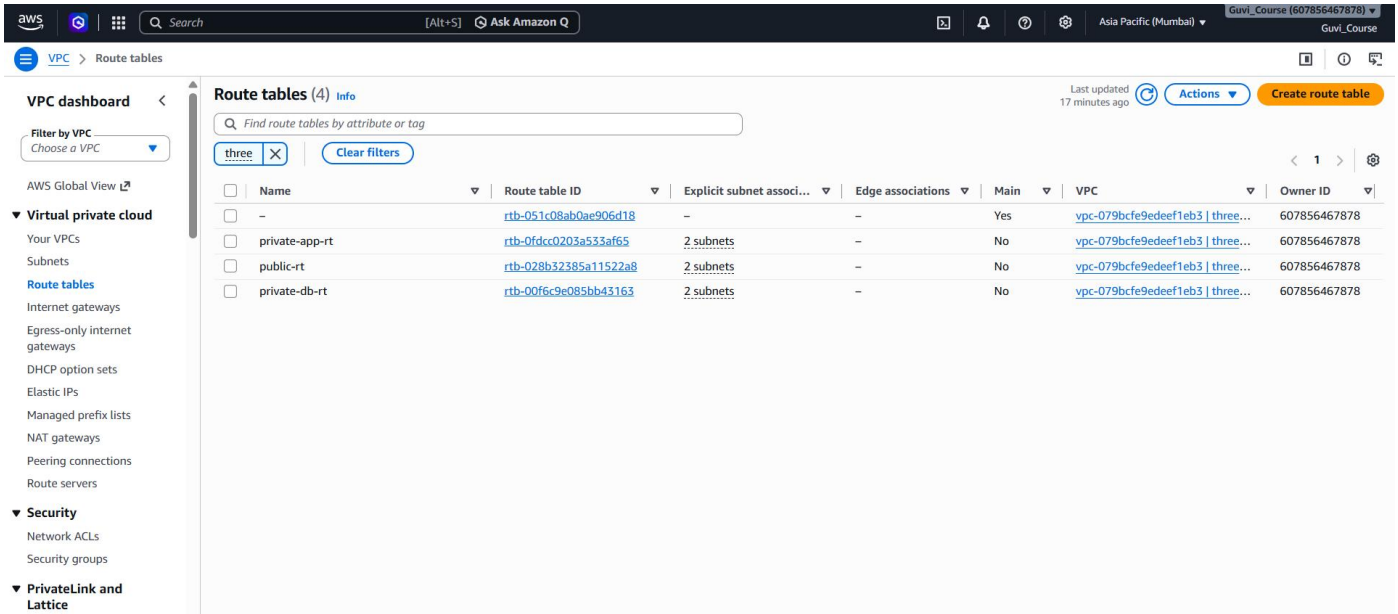
1. Created the internet gateway three-tier-igw and attached it to three-tier-vpc.
2. Allocated the Elastic IP 13.203.0.161.
3. Created the NAT gateway three-tier-nat in public-web-subnet-1a as a zonal, public NAT gateway using the Elastic IP.

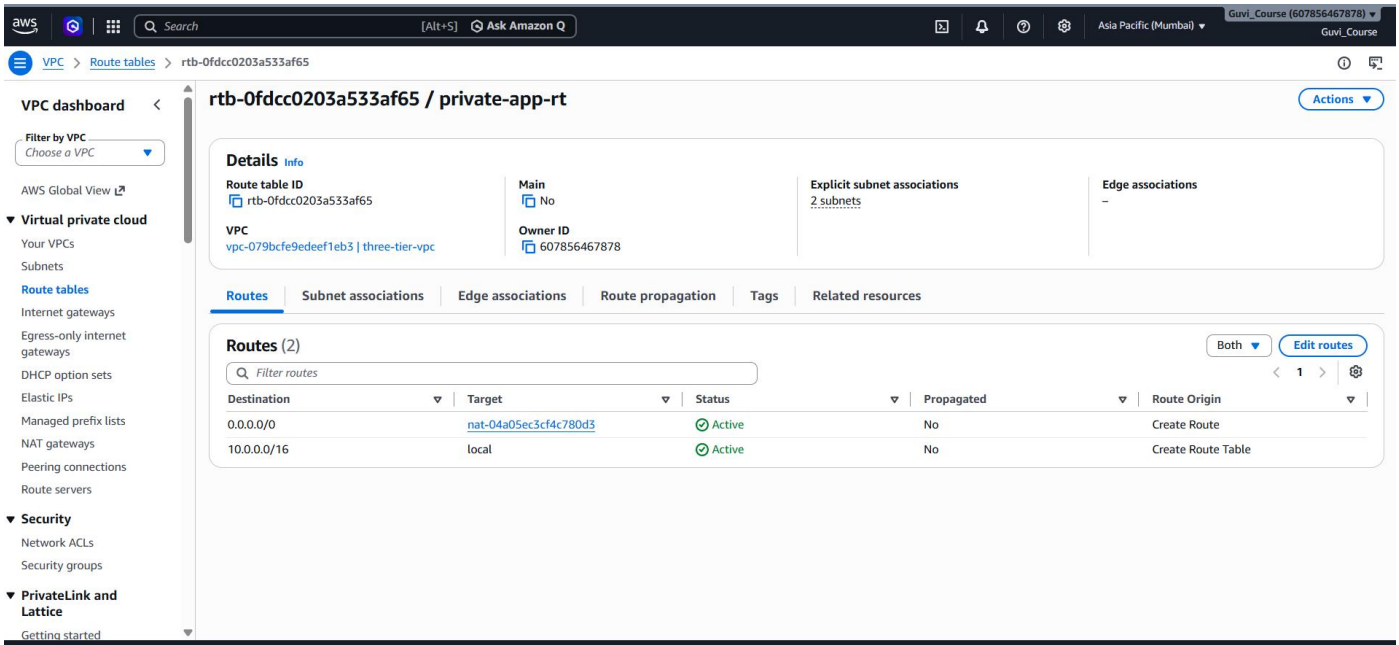


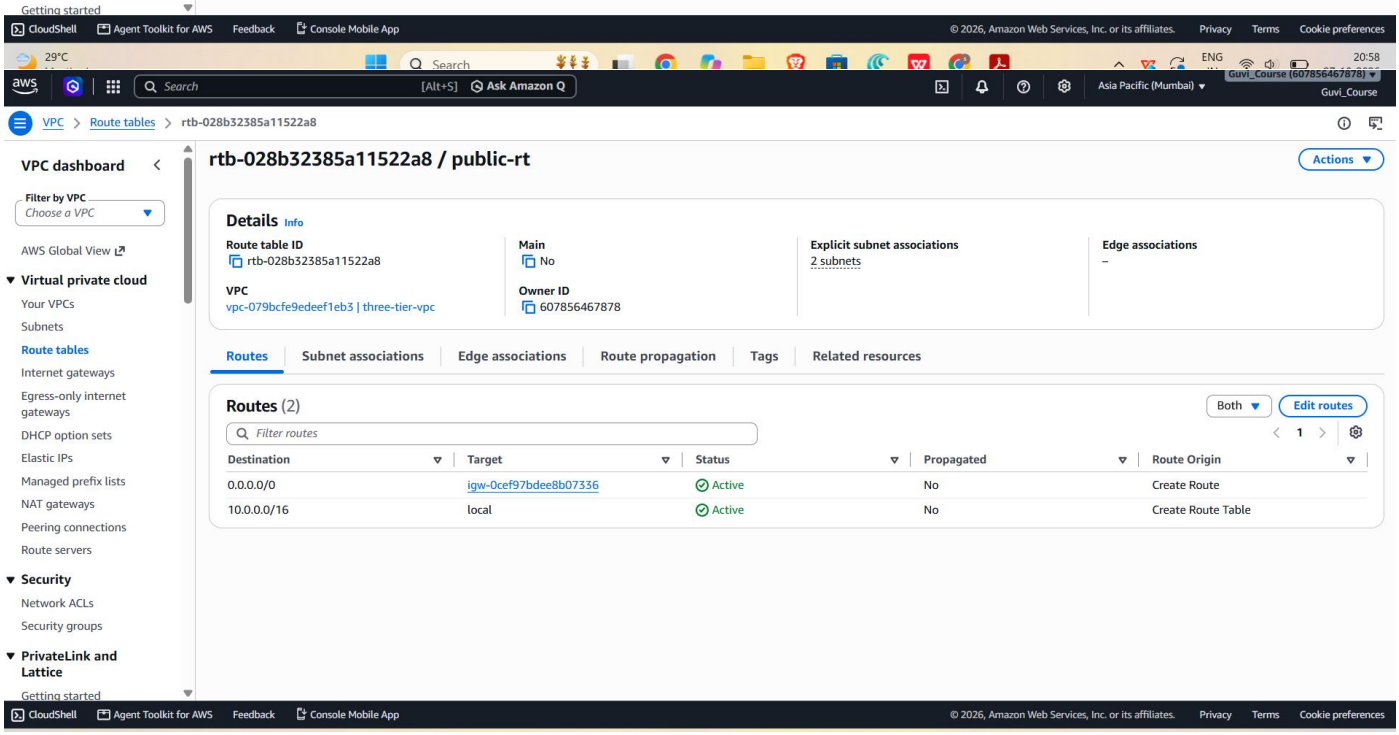


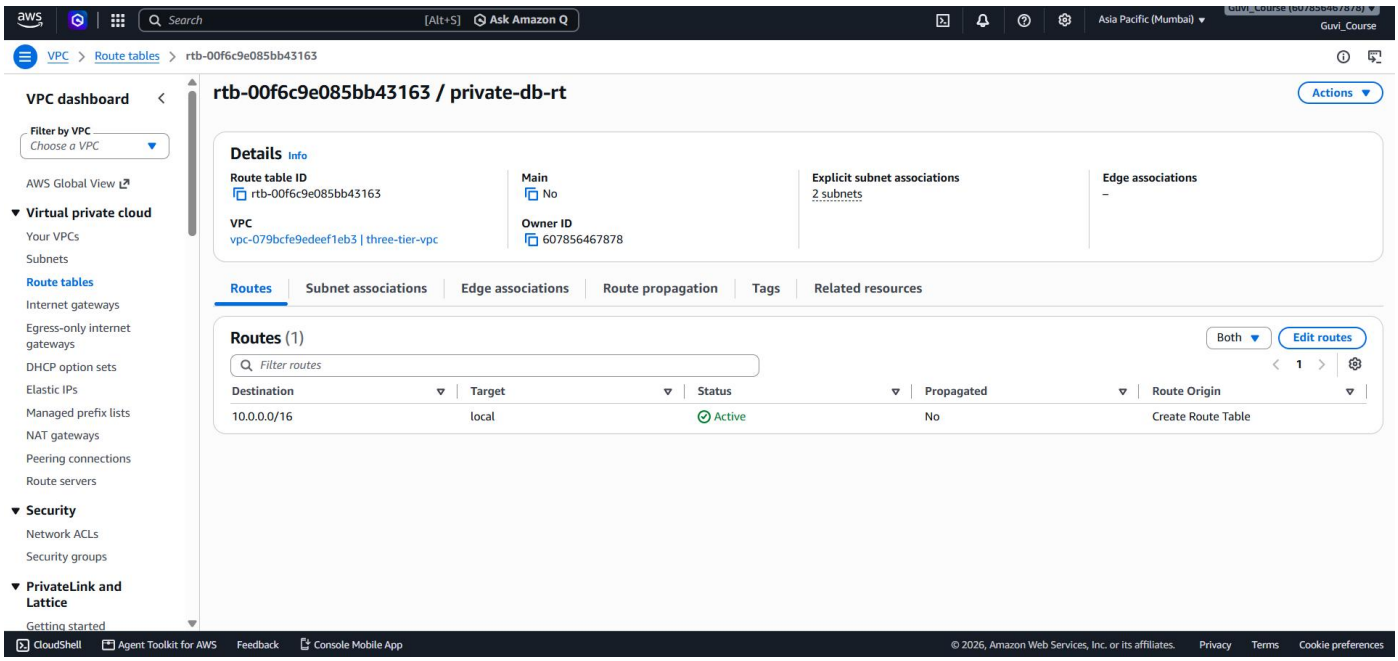
Route tables:

Route table	Routes	Associated subnets
public-rt	10.0.0.0/16 local; 0.0.0.0/0 → internet gateway	public-web-subnet-1a, public-web-subnet-1b
private-app-rt	10.0.0.0/16 local; 0.0.0.0/0 → NAT gateway	private-app-subnet-1a, private-app-subnet-1b
private-db-rt	10.0.0.0/16 local only (no internet route)	private-db-subnet-1a, private-db-subnet-1b




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Actions
Details Info
Route table ID
rtb-0fdcc0203a533af65
Main
No
Explicit subnet associations
2 subnets
Edge associations
–
VPC
vpc-079bcfe9edeef1eb3 | three-tier-vpc
Owner ID
607856467878
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (2)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0
nat-04a05ec3cf4c780d3
Active
No
Create Route
10.0.0.0/16
local
Active
No
Create Route Table
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No
Explicit subnet associations
2 subnets
Edge associations
–
VPC
vpc-079bcfe9edeef1eb3 | three-tier-vpc
Owner ID
607856467878
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (2)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0
igw-0cef97bdee8b07336
Active
No
Create Route
10.0.0.0/16
local
Active
No
Create Route Table
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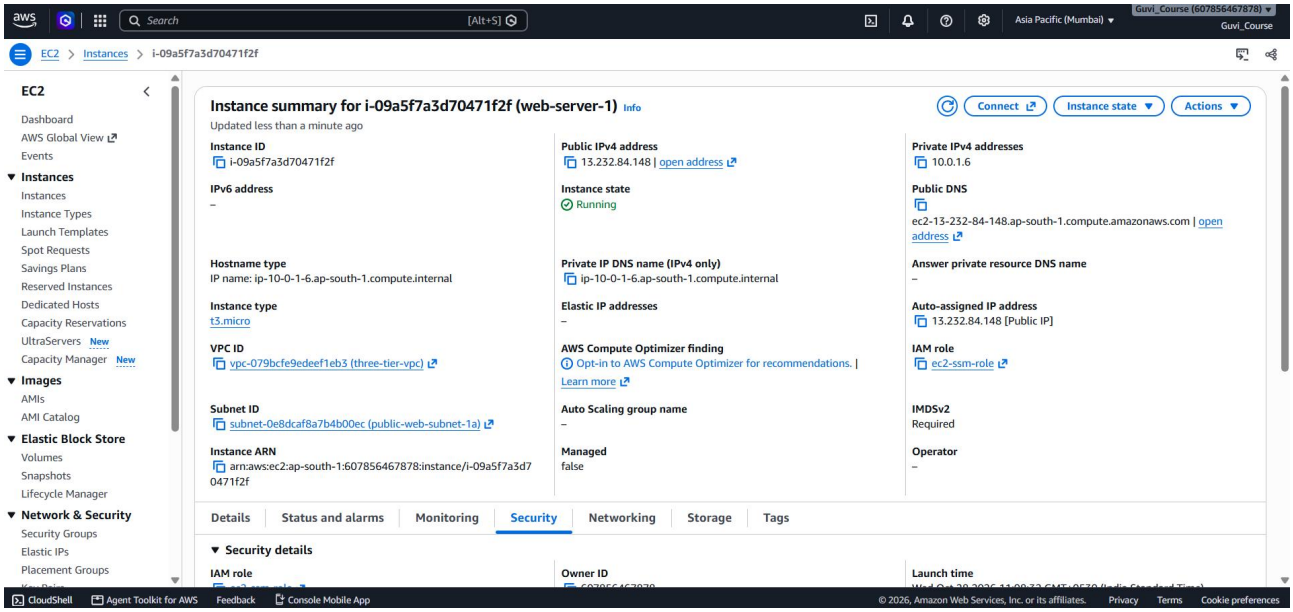
3. Task 2: Web tier:

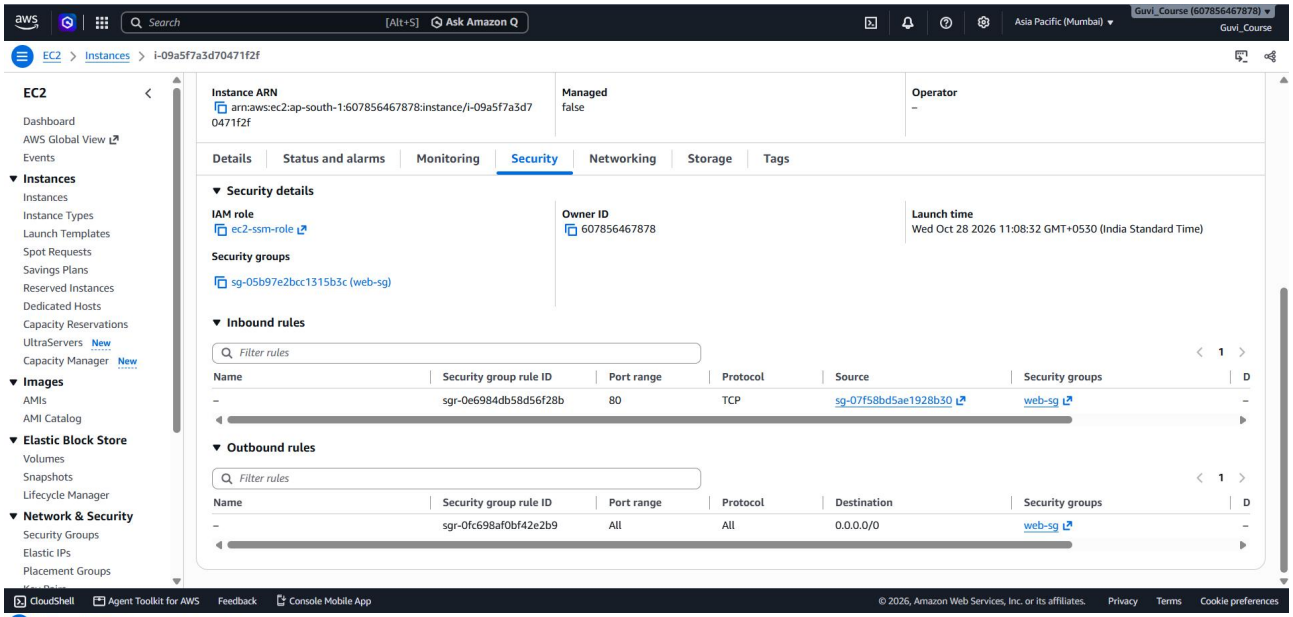
web-server-1 was launched from the Ubuntu Server 24.04 LTS AMI (t3.micro) in public-web-subnet-1a with a public IP, the web-sg security group and the ec2-ssm-role instance profile. No key pair was used; access was through Session Manager.

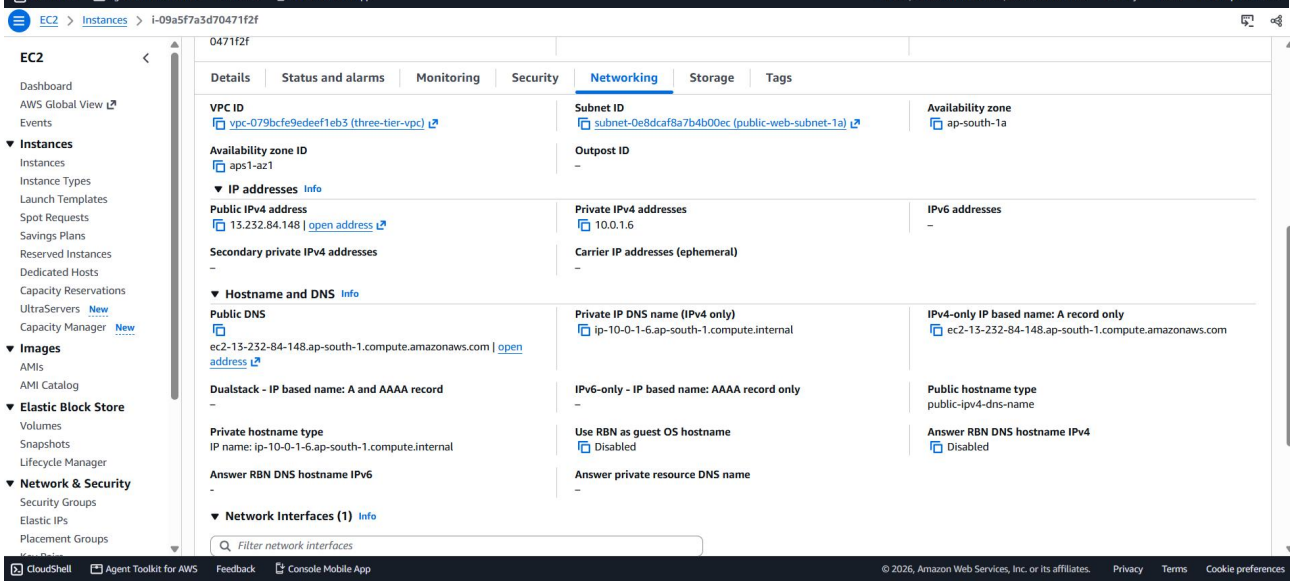
Commands:

```
sudo apt update
sudo apt install nginx git -y
git clone https://github.com/Vennilavanguvi/Three-Tier-Architecture.git
sudo cp Three-Tier-Architecture/index.html /usr/share/nginx/html/index.html
curl localhost
```

The curl command returned the HTML of the Feedback Form page, confirming that Nginx serves the provided file. A second web server (web-server-2) was created in public-web-subnet-1b from an AMI of the configured web-server-1 (web-server-ami), so it started with Nginx, the page and the proxy configuration already in place.


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EC2 > Instances > i-09a5f7a3d70471f2f
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Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Instance summary for i-09a5f7a3d70471f2f (web-server-1) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-09a5f7a3d70471f2f
Public IPv4 address
13.232.84.148 | open address
Private IPv4 addresses
10.0.1.6
IPv6 address
–
Instance state
Running
Public DNS
ec2-13-232-84-148.ap-south-1.compute.amazonaws.com | open address
Hostname type
IP name: ip-10-0-1-6.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-10-0-1-6.ap-south-1.compute.internal
Answer private resource DNS name
–
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
13.232.84.148 [Public IP]
VPC ID
vpc-079bcfe9edeef1eb3 (three-tier-vpc)
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role
ec2-ssm-role
Subnet ID
subnet-0e8dcaf8a7b4b00ec (public-web-subnet-1a)
Auto Scaling group name
–
IMDSv2
Required
Instance ARN
arn:aws:ec2:ap-south-1:607856467878:instance/i-09a5f7a3d70471f2f
Managed
false
Operator
–
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Security details
IAM role
Owner ID
Launch time
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Instance ARN
arn:aws:ec2:ap-south-1:607856467878:instance/i-09a5f7a3d70471f2f
Managed
false
Operator
–
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Security details
IAM role
ec2-ssm-role
Owner ID
607856467878
Launch time
Wed Oct 28 2026 11:08:32 GMT+0530 (India Standard Time)
Security groups
sg-05b97e2bcc1315b3c (web-sg)
Inbound rules
Filter rules
Name
Security group rule ID
Port range
Protocol
Source
Security groups
–
sgr-0e6984db58d56f28b
80
TCP
sg-07f58bd5ae1928b30
web-sg
Outbound rules
Filter rules
Name
Security group rule ID
Port range
Protocol
Destination
Security groups
–
sgr-0fc698af0bf42e2b9
All
All
0.0.0.0/0
web-sg
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EC2 > Instances > i-09a5f7a3d70471f2f
0471f2f
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
VPC ID
vpc-079bcfe9edeef1eb3 (three-tier-vpc)
Subnet ID
subnet-0e8dcaf8a7b4b00ec (public-web-subnet-1a)
Availability zone
ap-south-1a
Availability zone ID
aps1-az1
Outpost ID
–
IP addresses Info
Public IPv4 address
13.232.84.148 | open address
Private IPv4 addresses
10.0.1.6
IPv6 addresses
–
Secondary private IPv4 addresses
–
Carrier IP addresses (ephemeral)
–
Hostname and DNS Info
Public DNS
ec2-13-232-84-148.ap-south-1.compute.amazonaws.com | open address
Private IP DNS name (IPv4 only)
ip-10-0-1-6.ap-south-1.compute.internal
IPv4-only IP based name: A record only
ec2-13-232-84-148.ap-south-1.compute.amazonaws.com
Dualstack - IP based name: A and AAAA record
–
IPv6-only - IP based name: AAAA record only
–
Public hostname type
public-ipv4-dns-name
Private hostname type
IP name: ip-10-0-1-6.ap-south-1.compute.internal
Use RBN as guest OS hostname
Disabled
Answer RBN DNS hostname IPv4
Disabled
Answer RBN DNS hostname IPv6
-
Answer private resource DNS name
–
Network Interfaces (1) Info
Filter network interfaces
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Session ID: root-sdyuud3zvadflk7zsuui6ea89e

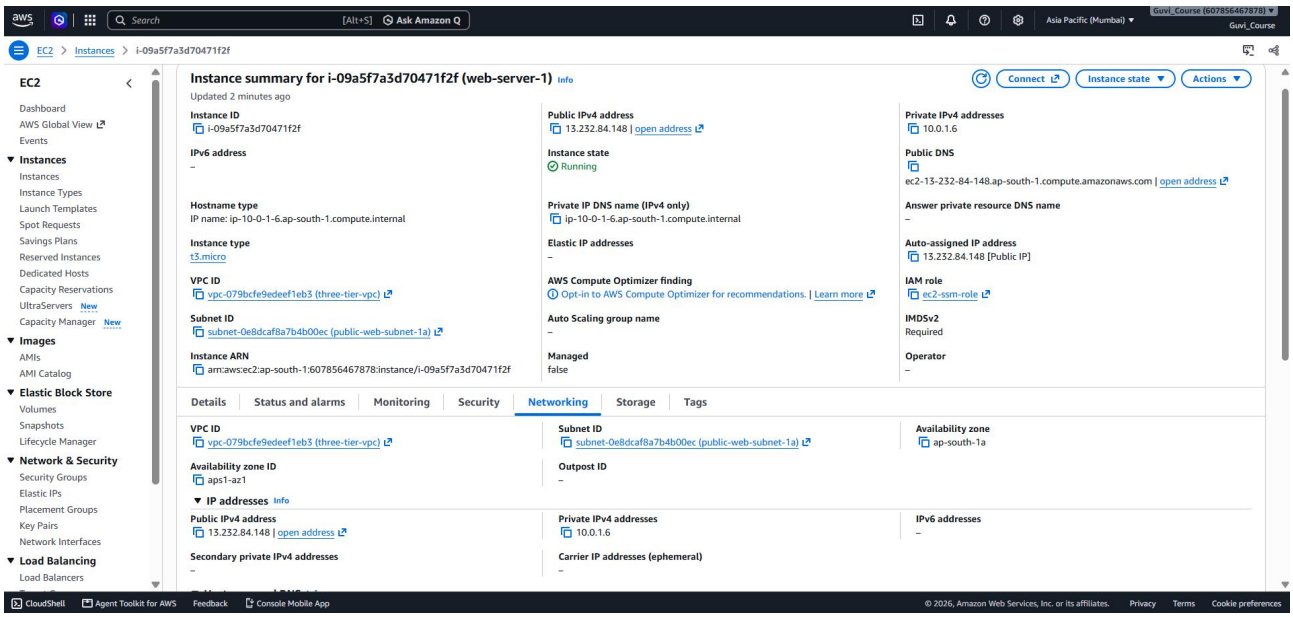
Shortcuts

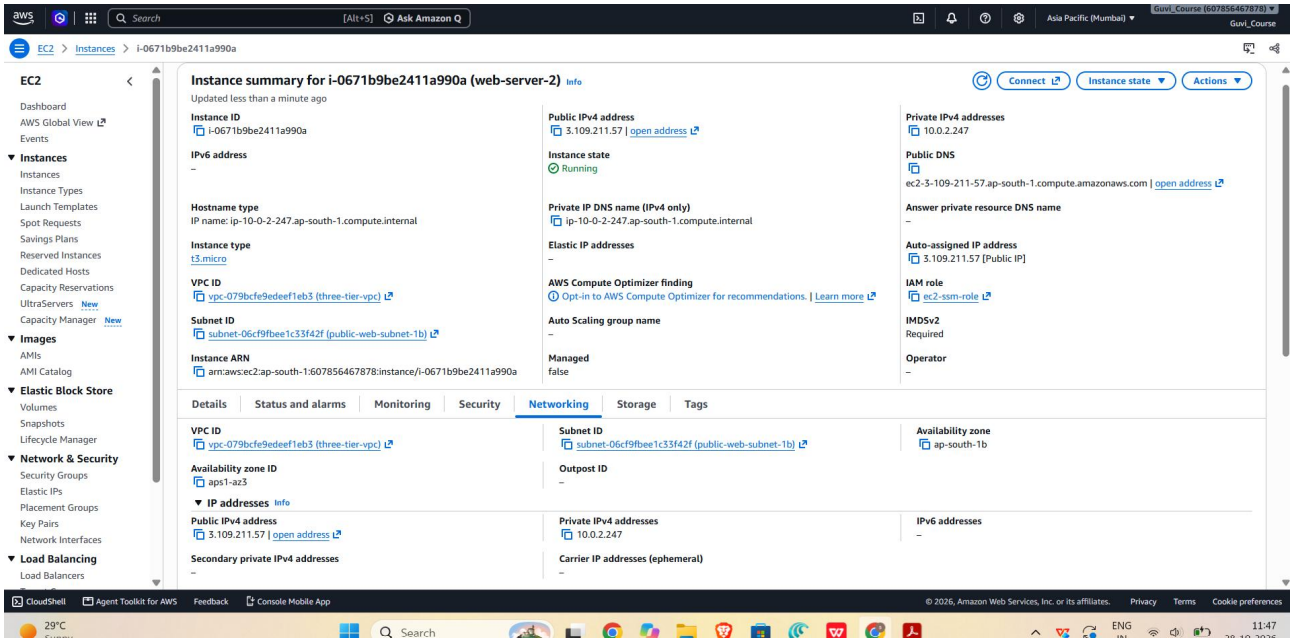
Instance ID: i-09a5f7a3d70471f2f (web-server-1)

Terminate session

```
$ curl localhost
<!DOCTYPE html>
<html lang='en'>
<head>
  <meta charset='UTF-8'>
  <title>Feedback Form</title>
  <style>
    body { font-family: Arial, sans-serif; max-width: 500px; margin: 60px auto; }
    input, textarea { width:100%; padding:10px; margin:8px 0; box-sizing:border-box; }
    button { background:#2563EB; color:#fff; padding:12px 28px; border:none; cursor:pointer; }
    #msg { margin-top:12px; font-weight:bold; }
  </style>
</head>
<body>
  <h2>Feedback Form</h2>
  <input id='name'     placeholder='Your Name' />
  <input id='email'    placeholder='Your Email' type='email' />
  <textarea id='feedback' rows='4' placeholder='Your Feedback'></textarea>
  <button onclick='submit()'>Submit</button>
  <p id='msg'></p>
  <script>
    function submit() {
      const name     = document.getElementById('name').value;
      const email    = document.getElementById('email').value;
      const feedback = document.getElementById('feedback').value;
      fetch('/api/submit', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ name, email, feedback })
      })
      .then(r => r.json())
      .then(d => { document.getElementById('msg').innerText = d.message; })
      .catch(() => { document.getElementById('msg').innerText = 'Error submitting.'; });
    }
  </script>
</body>
</html>
$
```

Web server 1 and web server 2 in different availability Zones:





4. Task 3: Application tier (Backend API)

app-server-1 was launched in private-app-subnet-1a with no public IP, the app-sg security group and the ec2-ssm-role instance profile. It was reached through Session Manager. Outbound internet access (apt, pip) goes through the NAT gateway, which was confirmed with a curl request to archive.ubuntu.com (HTTP 200).

Commands:

```
sudo su - ubuntu
curl -I https://archive.ubuntu.com        # NAT gateway check, HTTP 200
sudo apt update
sudo apt install python3-pip python3-venv python3-full default-mysql-client -y
git clone https://github.com/Vennilavanguvi/Three-Tier-Architecture.git
cp Three-Tier-Architecture/app.py ~/app.py
python3 -m venv venv && source venv/bin/activate
pip install Flask PyMySQL gunicorn
nohup python3 ~/app.py > ~/app.log 2>&1 &
curl localhost:5000                        # {"status":"Flask is running"}
```

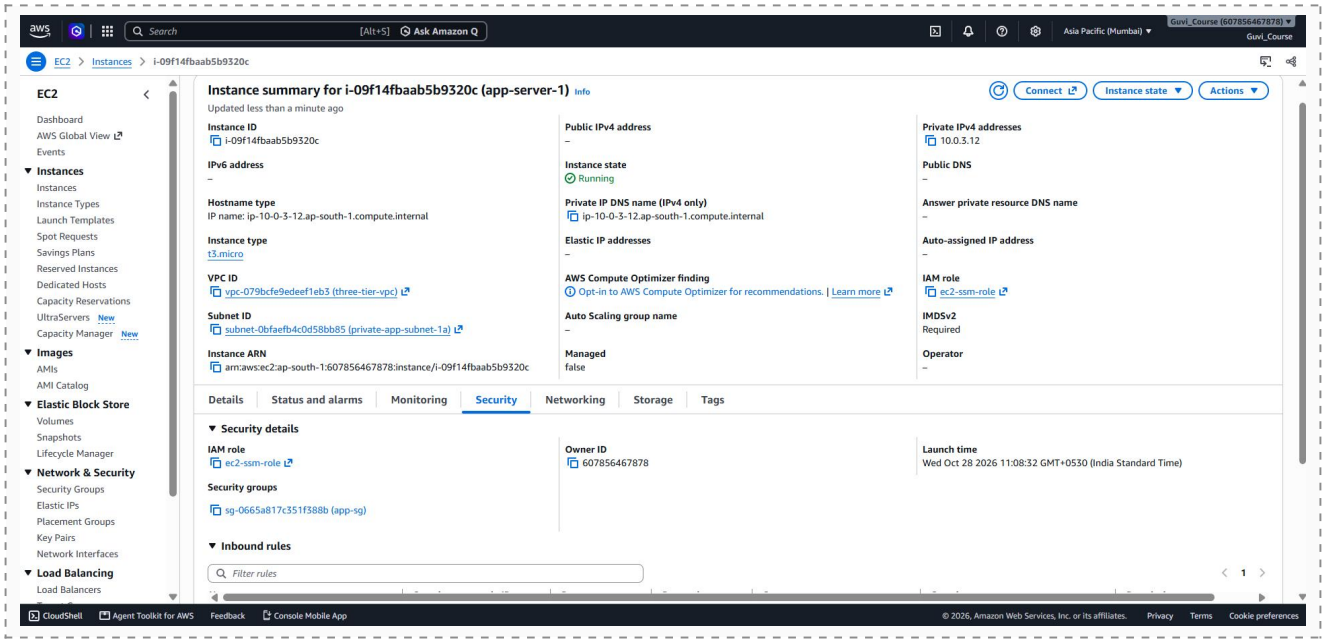
To keep Flask running after logout and reboot, it was configured as a systemd service (flask-app.service, Restart=always). An AMI (app-server-ami) was then created and used to launch app-server-2 in private-app-subnet-1b.

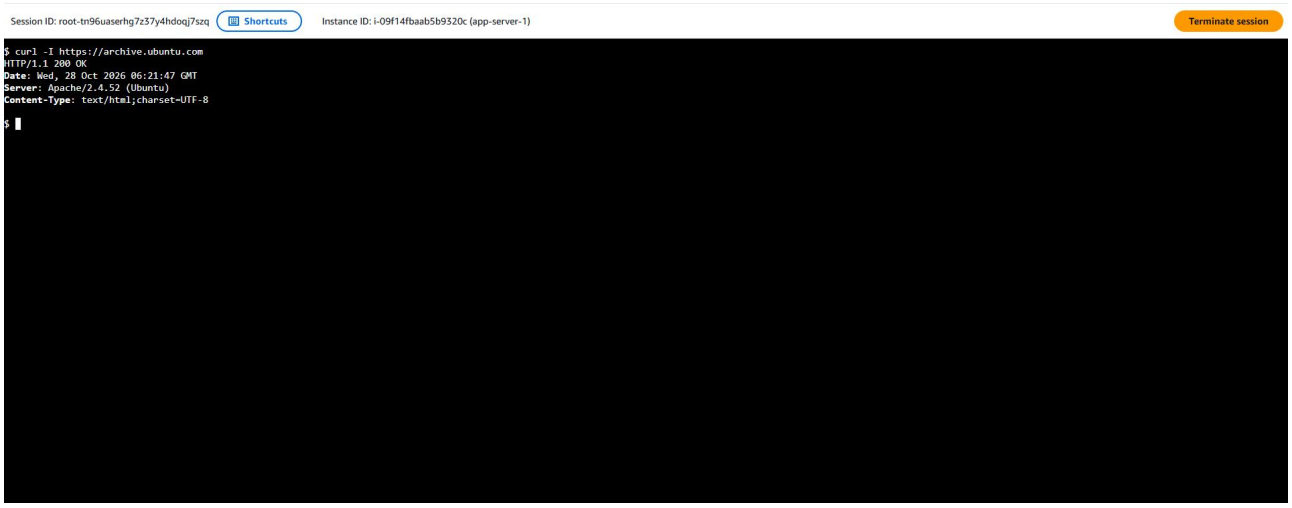
systemd service file (/etc/systemd/system/flask-app.service)

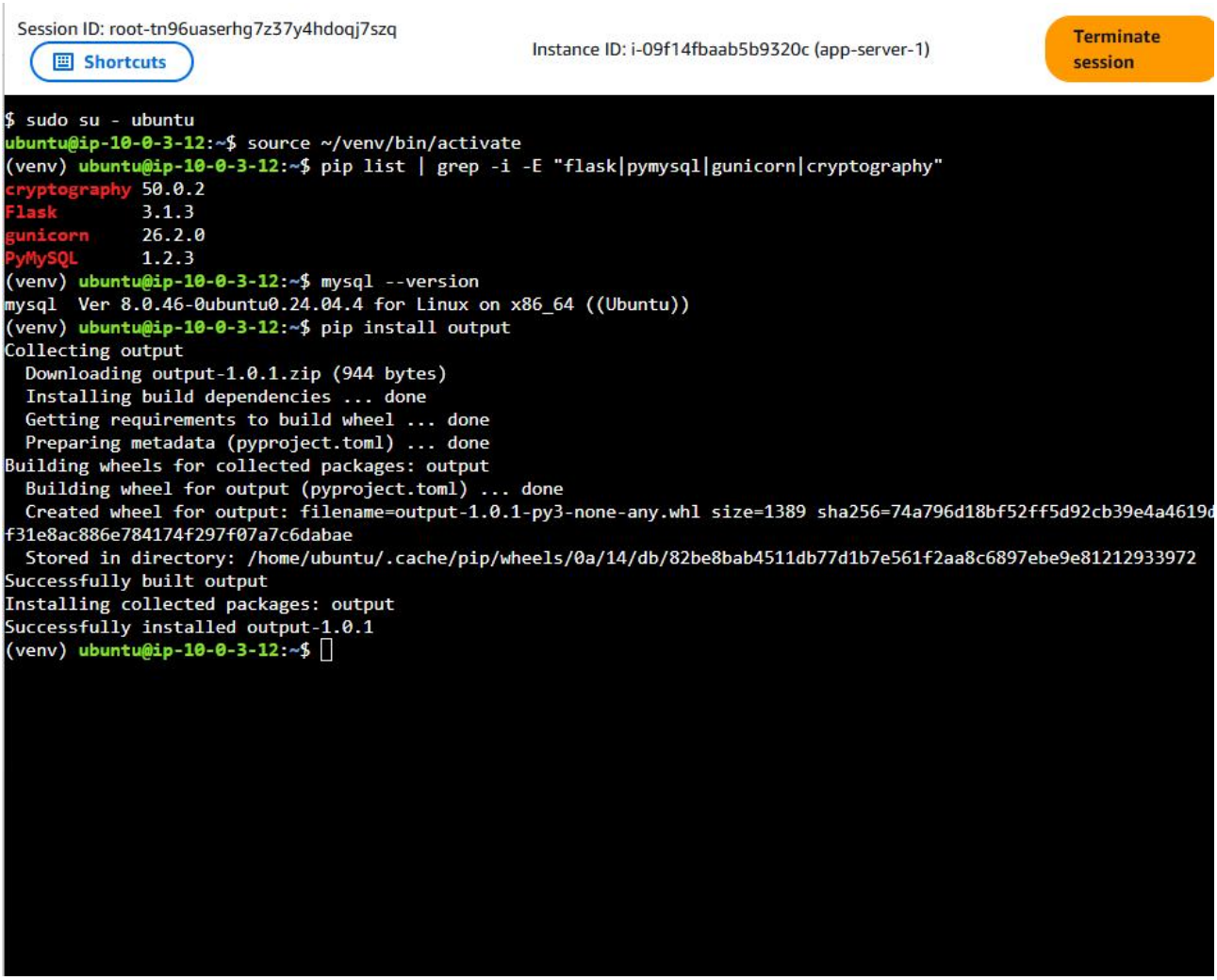
```
[Unit]
Description=Flask Feedback App
After=network.target

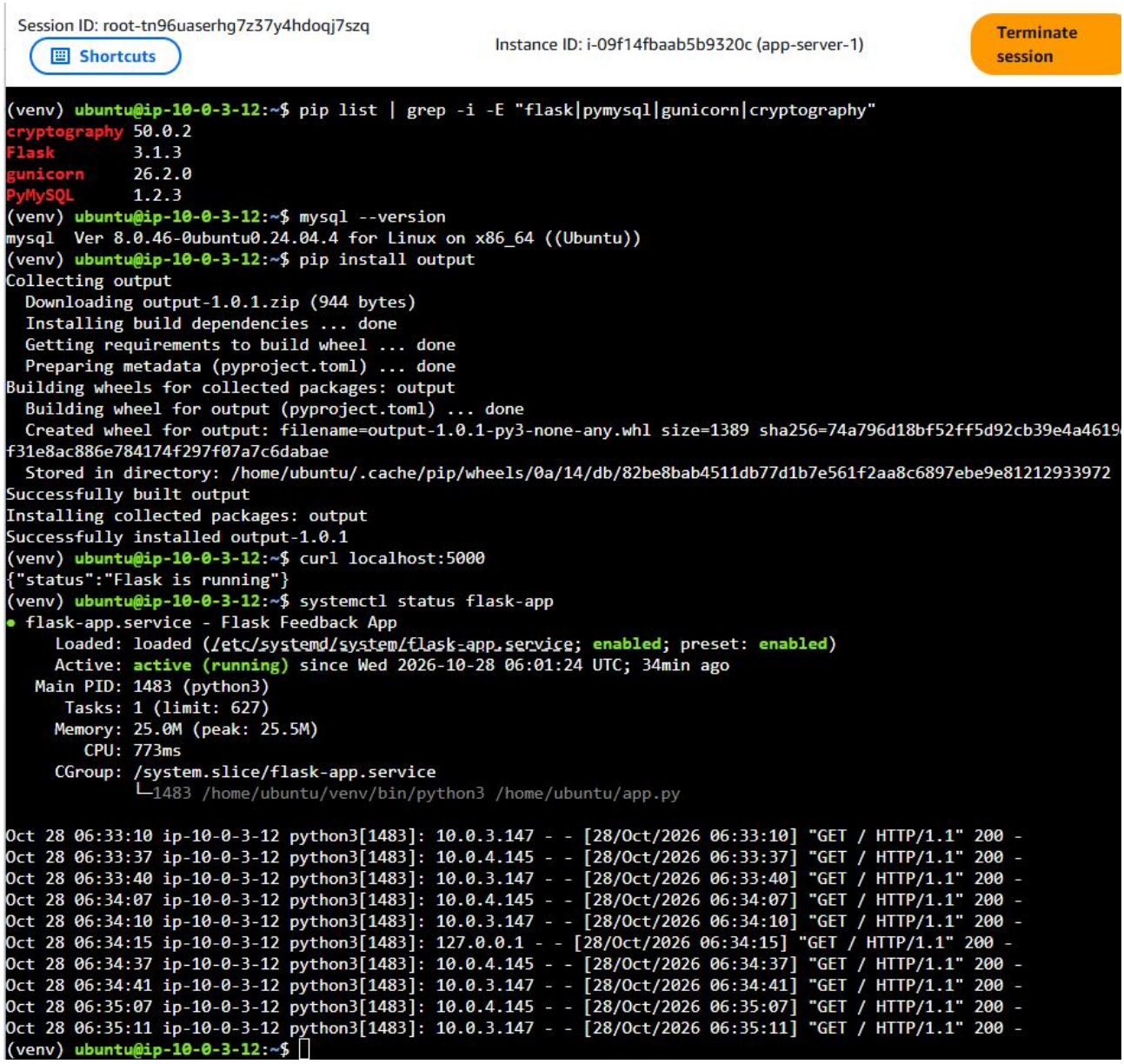
[Service]
User=ubuntu
WorkingDirectory=/home/ubuntu
ExecStart=/home/ubuntu/venv/bin/python3 /home/ubuntu/app.py
Restart=always

[Install]
WantedBy=multi-user.target
```

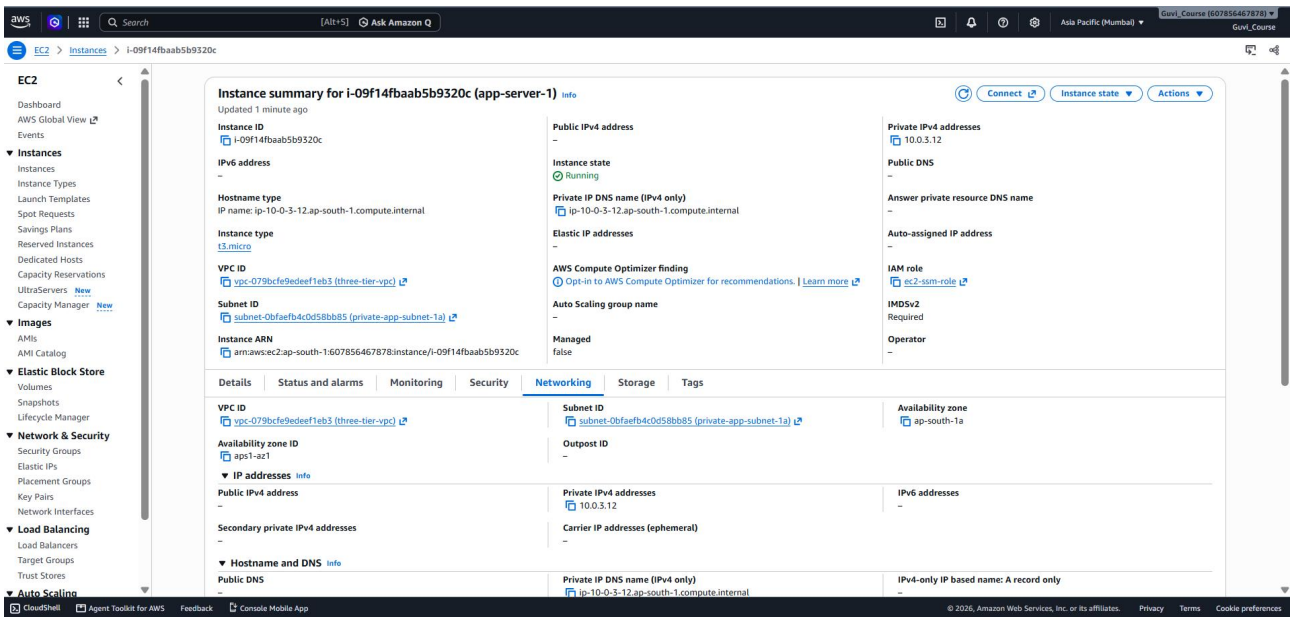


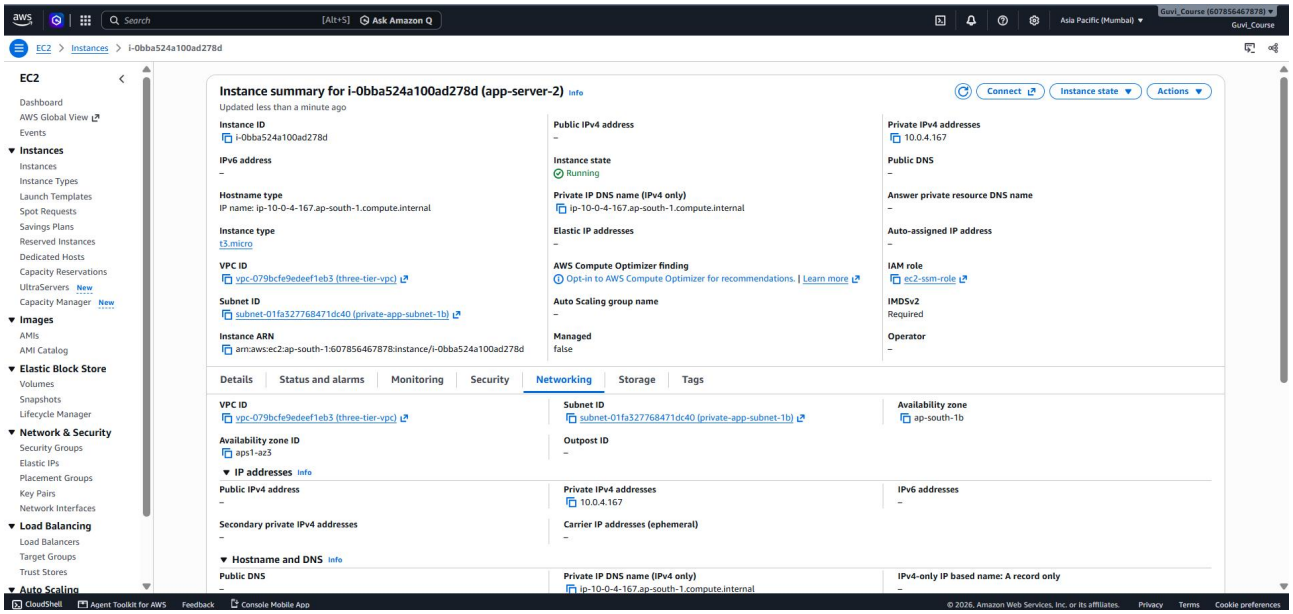



Session ID: root-tn96uaserhg7z37y4hdoqj7szq
Shortcuts
Instance ID: i-09f14fbaab5b9320c (app-server-1)
Terminate session
$ sudo su - ubuntu
ubuntu@ip-10-0-3-12:~$ source ~/venv/bin/activate
(venv) ubuntu@ip-10-0-3-12:~$ pip list | grep -i -E "flask|pymysql|gunicorn|cryptography"
cryptography 50.0.2
Flask 3.1.3
gunicorn 26.2.0
PyMySQL 1.2.3
(venv) ubuntu@ip-10-0-3-12:~$ mysql --version
mysql Ver 8.0.46-0ubuntu0.24.04.4 for Linux on x86_64 ((Ubuntu))
(venv) ubuntu@ip-10-0-3-12:~$ pip install output
Collecting output
Downloading output-1.0.1.zip (944 bytes)
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: output
Building wheel for output (pyproject.toml) ... done
Created wheel for output: filename=output-1.0.1-py3-none-any.whl size=1389 sha256=74a796d18bf52ff5d92cb39e4a4619d
f31e8ac886e784174f297f07a7c6dabae
Stored in directory: /home/ubuntu/.cache/pip/wheels/0a/14/db/82be8bab4511db77d1b7e561f2aa8c6897ebe9e81212933972
Successfully built output
Installing collected packages: output
Successfully installed output-1.0.1
(venv) ubuntu@ip-10-0-3-12:~$



app server 1 and app server 2 in different availability Zones:


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EC2 > Instances > i-09f14fbaab5b9320c
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Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Instance summary for i-09f14fbaab5b9320c (app-server-1) Info
Updated 1 minute ago
Connect
Instance state
Actions
Instance ID
i-09f14fbaab5b9320c
Public IPv4 address
–
Private IPv4 addresses
10.0.3.12
IPv6 address
–
Instance state
Running
Public DNS
–
Hostname type
IP name: ip-10-0-3-12.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-10-0-3-12.ap-south-1.compute.internal
Answer private resource DNS name
–
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
–
VPC ID
vpc-079bcfe9edeef1eb3 (three-tier-vpc)
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role
ec2-ssm-role
Subnet ID
subnet-0bfaefb4c0d58bb85 (private-app-subnet-1a)
Auto Scaling group name
–
IMDSv2
Required
Instance ARN
arn:aws:ec2:ap-south-1:607856467878:instance/i-09f14fbaab5b9320c
Managed
false
Operator
–
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
VPC ID
vpc-079bcfe9edeef1eb3 (three-tier-vpc)
Subnet ID
subnet-0bfaefb4c0d58bb85 (private-app-subnet-1a)
Availability zone
ap-south-1a
Availability zone ID
aps1-az1
Outpost ID
–
IP addresses Info
Public IPv4 address
–
Private IPv4 addresses
10.0.3.12
IPv6 addresses
–
Secondary private IPv4 addresses
–
Carrier IP addresses (ephemeral)
–
Hostname and DNS Info
Public DNS
–
Private IP DNS name (IPv4 only)
ip-10-0-3-12.ap-south-1.compute.internal
IPv4-only IP based name: A record only
–
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Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Instance summary for i-0bba524a100ad278d (app-server-2) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-0bba524a100ad278d
Public IPv4 address
–
Private IPv4 addresses
10.0.4.167
IPv6 address
–
Instance state
Running
Public DNS
–
Hostname type
IP name: ip-10-0-4-167.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-10-0-4-167.ap-south-1.compute.internal
Answer private resource DNS name
–
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
–
VPC ID
vpc-079bcfe9edeef1eb3 (three-tier-vpc)
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role
ec2-ssm-role
Subnet ID
subnet-01fa327768471dc40 (private-app-subnet-1b)
Auto Scaling group name
–
IMDSv2
Required
Instance ARN
arn:aws:ec2:ap-south-1:607856467878:instance/i-0bba524a100ad278d
Managed
false
Operator
–
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
VPC ID
vpc-079bcfe9edeef1eb3 (three-tier-vpc)
Subnet ID
subnet-01fa327768471dc40 (private-app-subnet-1b)
Availability zone
ap-south-1b
Availability zone ID
aps1-az3
Outpost ID
–
IP addresses Info
Public IPv4 address
–
Private IPv4 addresses
10.0.4.167
IPv6 addresses
–
Secondary private IPv4 addresses
–
Carrier IP addresses (ephemeral)
–
Hostname and DNS Info
Public DNS
–
Private IP DNS name (IPv4 only)
ip-10-0-4-167.ap-south-1.compute.internal
IPv4-only IP based name: A record only
–
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5. Task 4: DB tier

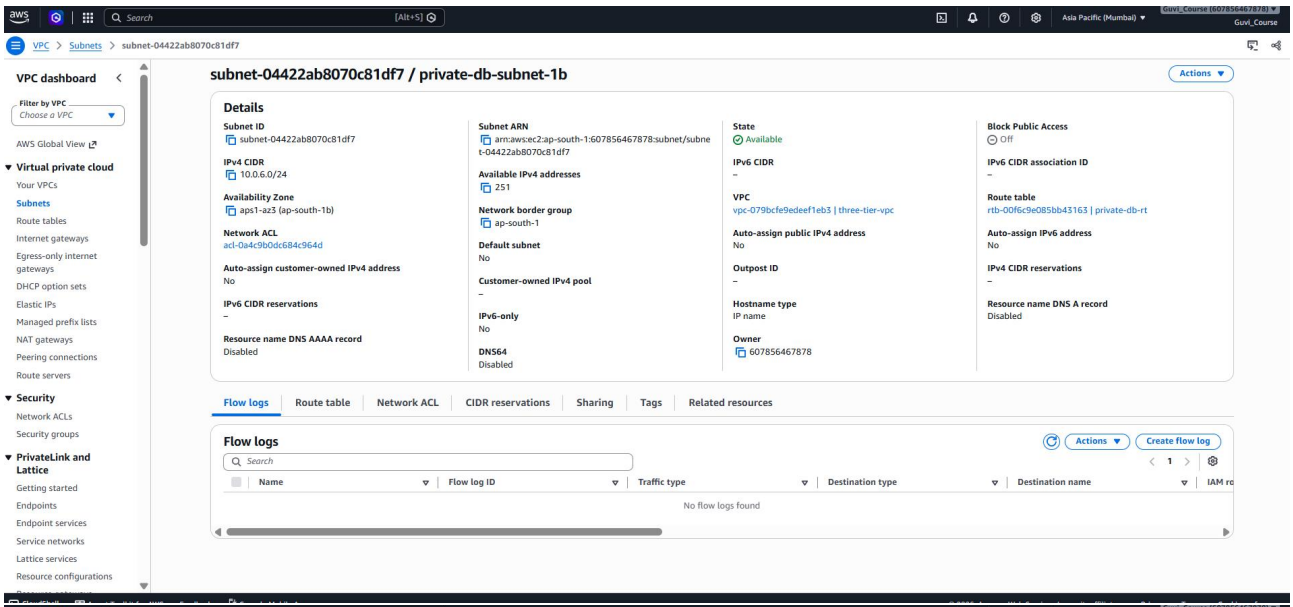
A DB subnet group (three-tier-db-subnet-group) containing only the two private database subnets was created, then the RDS MySQL instance was launched into it.

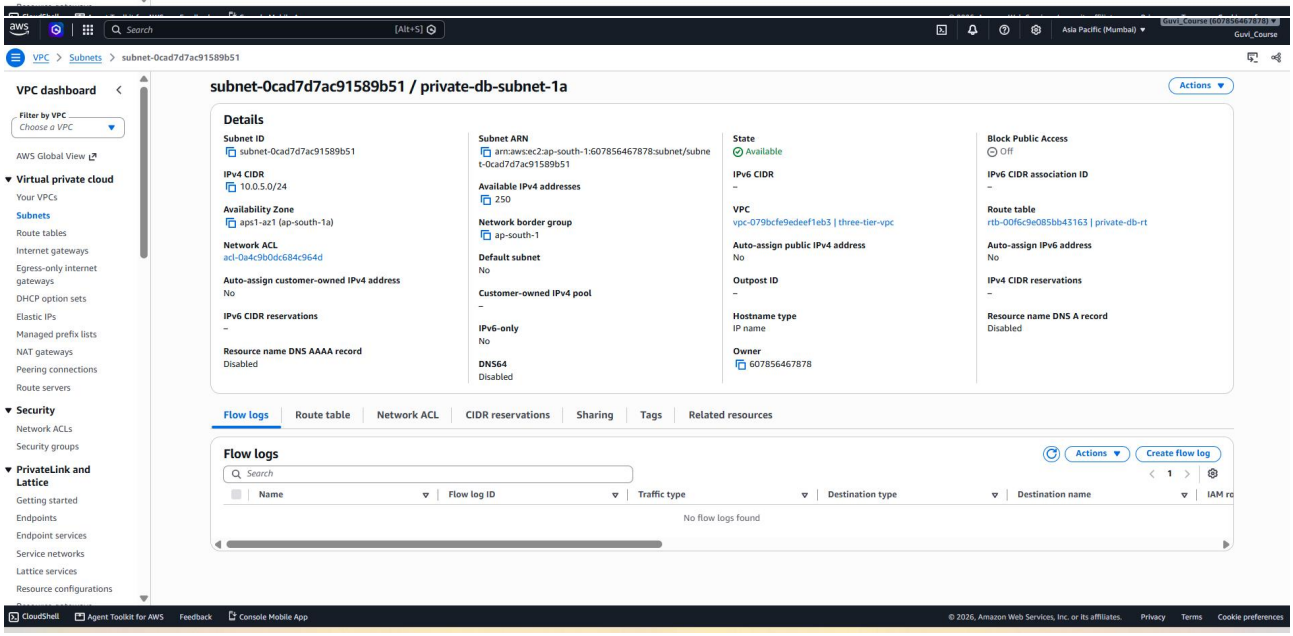
Setting	Value
Engine	MySQL Community 8.4.11
Template / deployment	Free tier, single-AZ
DB instance identifier	three-tier-db
Instance class / storage	db.t4g.micro, 20 GiB
Master username	admin (password stored)
VPC / subnet group	three-tier-vpc / three-tier-db-subnet-group (10.0.5.0/24, 10.0.6.0/24)
Public access	No
Security group	db-sg (MySQL 3306 from app-sg only)
Initial database name	database is created by the SQL below

SQL from DB.md, run from app-server-1

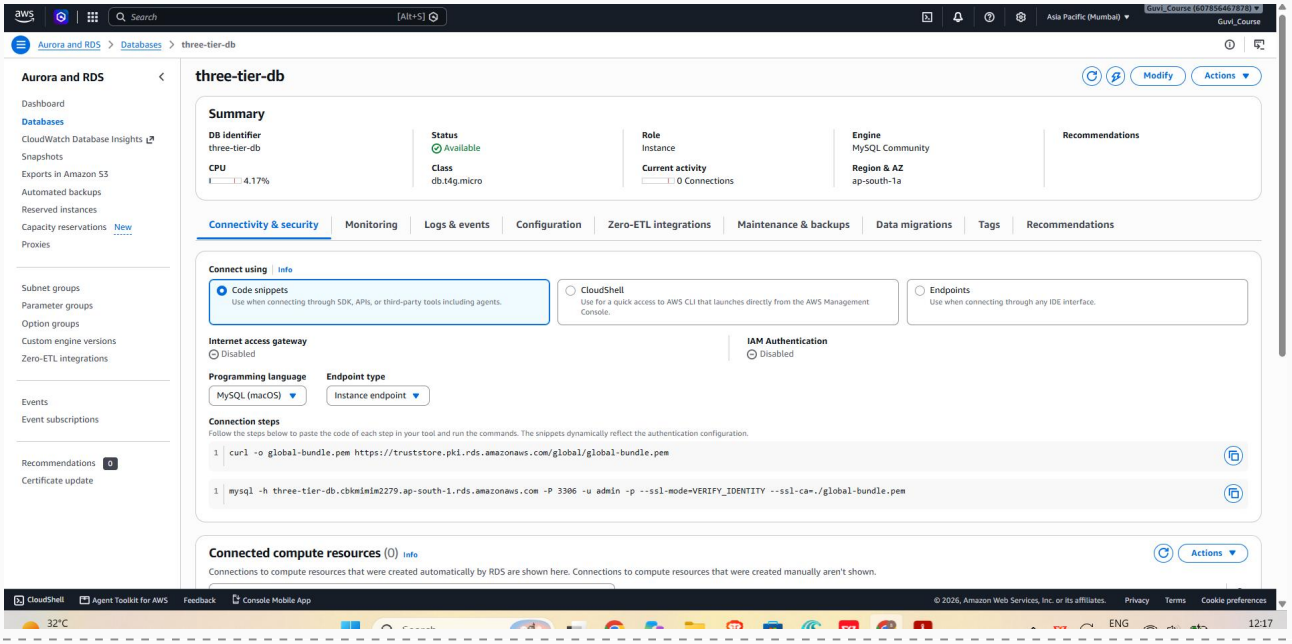
```
mysql -h three-tier-db.cbkmimim2279.ap-south-1.rds.amazonaws.com -u admin -p

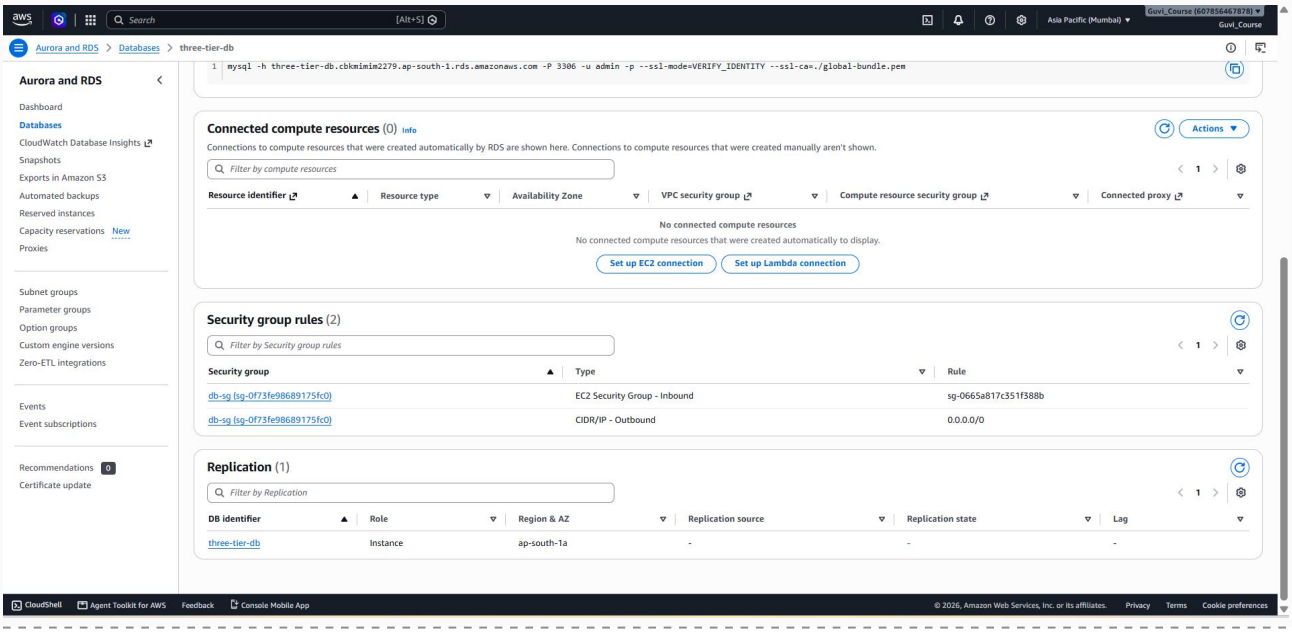
CREATE DATABASE feedbackdb;
USE feedbackdb;
CREATE TABLE feedback (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(100),
    feedback TEXT,
    created TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
SHOW TABLES;
DESCRIBE feedback;
```

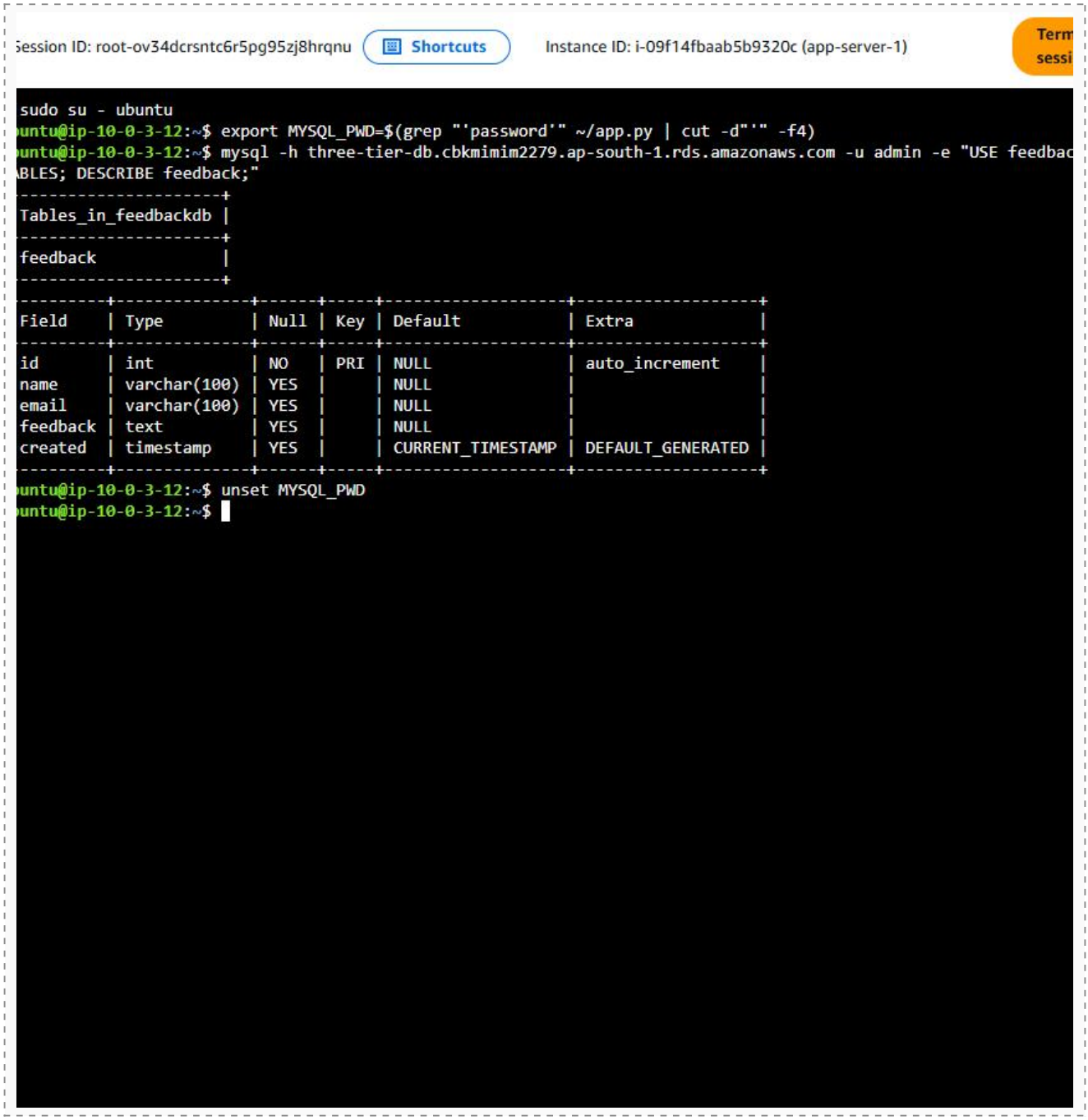

VPC > Subnets > subnet-04422ab8070c81df7
VPC dashboard
subnet-04422ab8070c81df7 / private-db-subnet-1b
Actions
Details
Subnet ID
subnet-04422ab8070c81df7
IPv4 CIDR
10.0.6.0/24
Availability Zone
aps1-az3 (ap-south-1b)
Network ACL
acl-0a4c9b0dc684c964d
Auto-assign customer-owned IPv4 address
No
IPv6 CIDR reservations
–
Resource name DNS AAAA record
Disabled
Subnet ARN
arn:aws:ec2:ap-south-1:607856467878:subnet/subnet-04422ab8070c81df7
Available IPv4 addresses
251
Network border group
ap-south-1
Default subnet
No
Customer-owned IPv4 pool
–
IPv6-only
No
DNS64
Disabled
State
Available
IPv6 CIDR
–
VPC
vpc-079bcfe9edeef1eb3 | three-tier-vpc
Auto-assign public IPv4 address
No
Outpost ID
–
Hostname type
IP name
Owner
607856467878
Block Public Access
Off
IPv6 CIDR association ID
–
Route table
rtb-00f6c9e085bb43163 | private-db-rt
Auto-assign IPv6 address
No
IPv4 CIDR reservations
–
Resource name DNS A record
Disabled
Flow logs
Route table
Network ACL
CIDR reservations
Sharing
Tags
Related resources
Flow logs
Actions
Create flow log
Search
Name
Flow log ID
Traffic type
Destination type
Destination name
No flow logs found


VPC > Subnets > subnet-0cad7d7ac91589b51
VPC dashboard
subnet-0cad7d7ac91589b51 / private-db-subnet-1a
Actions
Details
Subnet ID
subnet-0cad7d7ac91589b51
IPv4 CIDR
10.0.5.0/24
Availability Zone
aps1-az1 (ap-south-1a)
Network ACL
acl-0a4c9b0dc684c964d
Auto-assign customer-owned IPv4 address
No
IPv6 CIDR reservations
–
Resource name DNS AAAA record
Disabled
Subnet ARN
arn:aws:ec2:ap-south-1:607856467878:subnet/subnet-0cad7d7ac91589b51
Available IPv4 addresses
250
Network border group
ap-south-1
Default subnet
No
Customer-owned IPv4 pool
–
IPv6-only
No
DNS64
Disabled
State
Available
IPv6 CIDR
–
VPC
vpc-079bcfe9edeef1eb3 | three-tier-vpc
Auto-assign public IPv4 address
No
Outpost ID
–
Hostname type
IP name
Owner
607856467878
Block Public Access
Off
IPv6 CIDR association ID
–
Route table
rtb-00f6c9e085bb43163 | private-db-rt
Auto-assign IPv6 address
No
IPv4 CIDR reservations
–
Resource name DNS A record
Disabled
Flow logs
Route table
Network ACL
CIDR reservations
Sharing
Tags
Related resources
Flow logs
Actions
Create flow log
Search
Name
Flow log ID
Traffic type
Destination type
Destination name
No flow logs found
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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RDS Connectivity & security tab: endpoint, Publicly accessible No, db-sg:



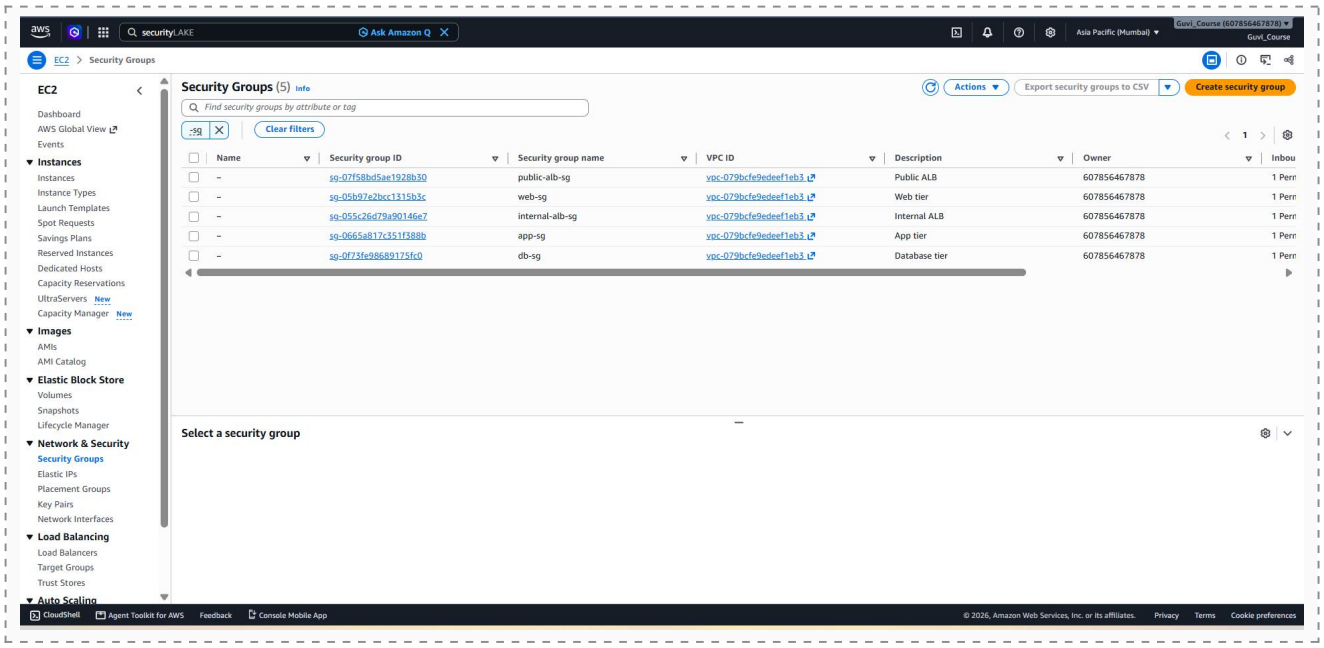


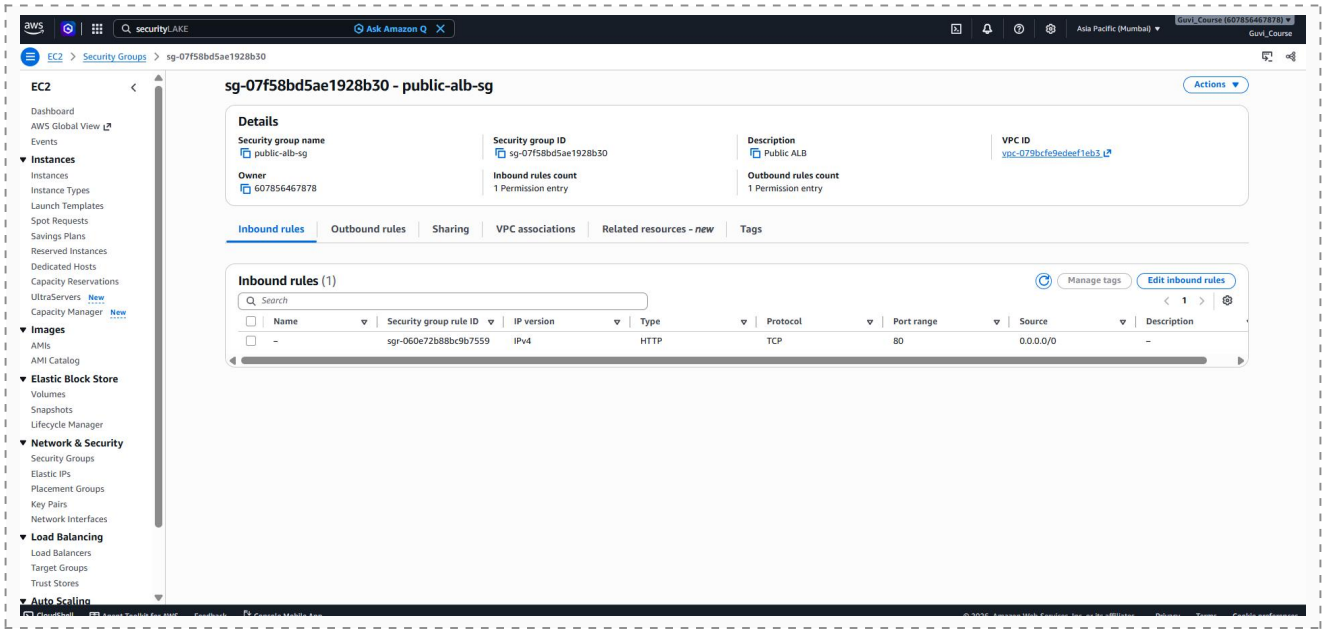

Session ID: root-ov34dcrsntc6r5pg95zj8hrqnu
Shortcuts
Instance ID: i-09f14fbaab5b9320c (app-server-1)
sudo su - ubuntu
ubuntu@ip-10-0-3-12:~$ export MYSQL_PWD=$(grep "'password'" ~/app.py | cut -d"'" -f4)
ubuntu@ip-10-0-3-12:~$ mysql -h three-tier-db.cbkmimim2279.ap-south-1.rds.amazonaws.com -u admin -e "USE feedbac
ABLES; DESCRIBE feedback;"
Tables_in_feedbackdb
feedback
Field | Type | Null | Key | Default | Extra
id | int | NO | PRI | NULL | auto_increment
name | varchar(100) | YES | | NULL |
email | varchar(100) | YES | | NULL |
feedback | text | YES | | NULL |
created | timestamp | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED
ubuntu@ip-10-0-3-12:~$ unset MYSQL_PWD
ubuntu@ip-10-0-3-12:~$

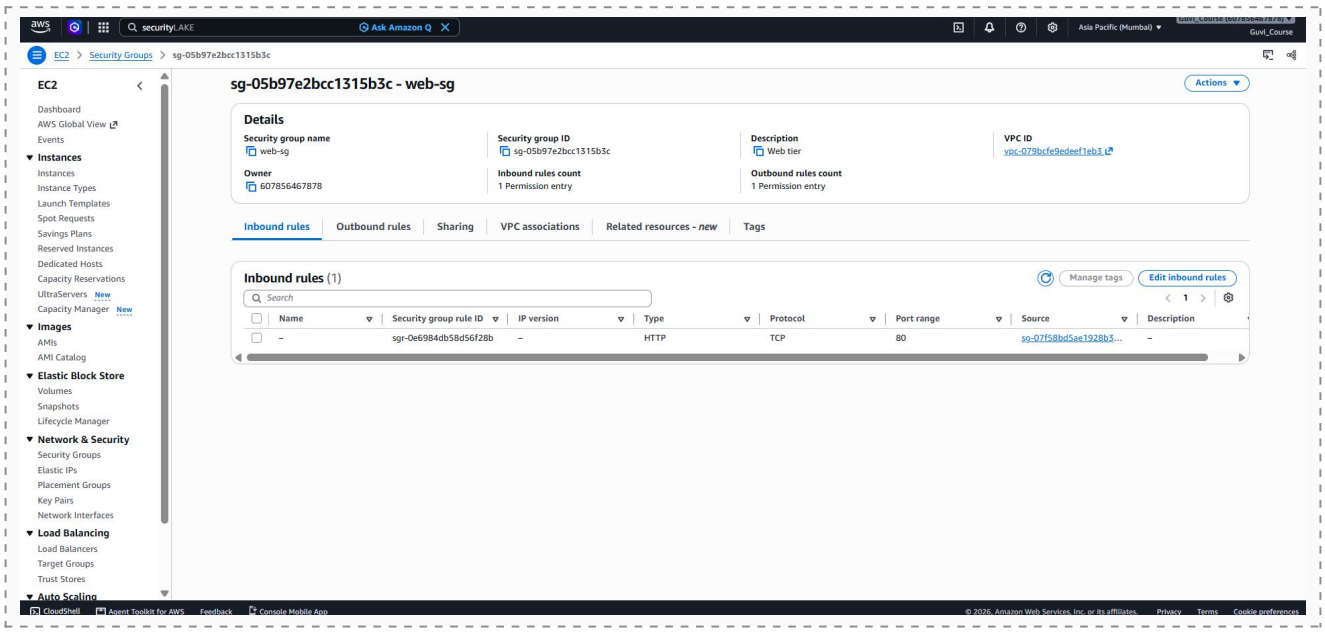
6. Task 5: Network and instance security

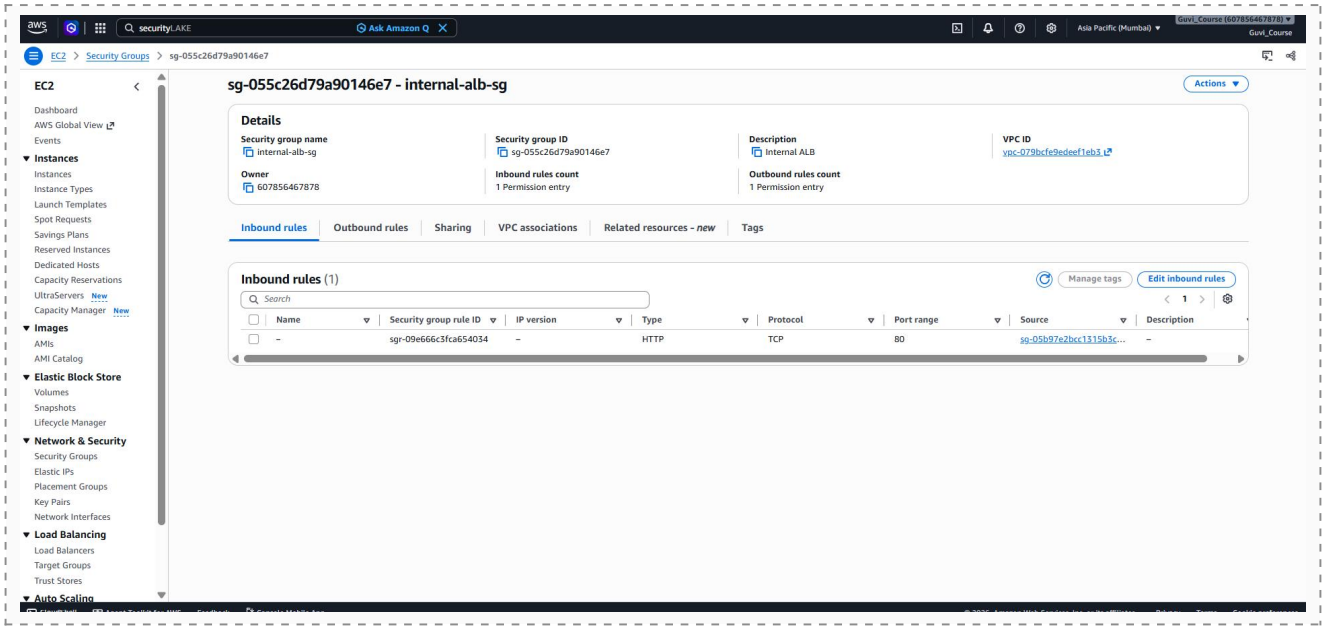
Each tier accepts traffic only from the tier in front of it. Security groups reference other security groups instead of IP ranges, and no SSH rule is open anywhere because administration is done through Session Manager.

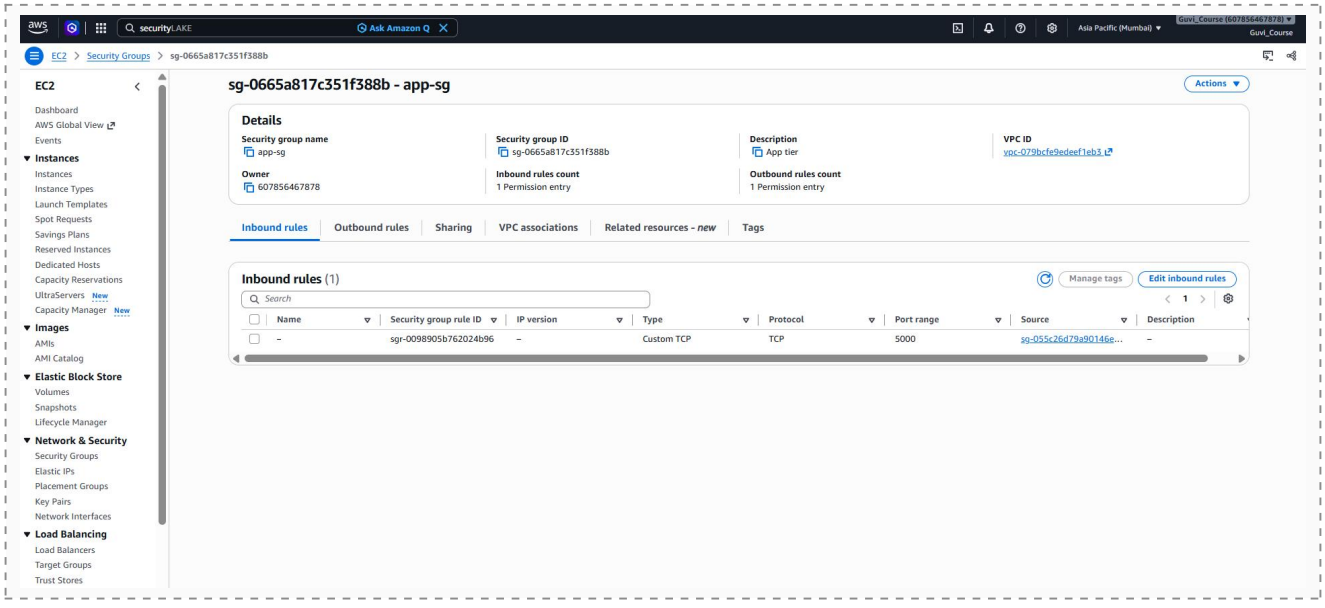
Security group	ID	Inbound rule	Source
public-alb-sg	sg-07f58bd5ae1928b30	HTTP 80	0.0.0.0/0
web-sg	sg-05b97e2bcc1315b3c	HTTP 80	public-alb-sg
internal-alb-sg	sg-055c26d79a90146e7	HTTP 80	web-sg
app-sg	sg-0665a817c351f388b	Custom TCP 5000	internal-alb-sg
db-sg	sg-0f73fe98689175fc0	MySQL/Aurora 3306	app-sg

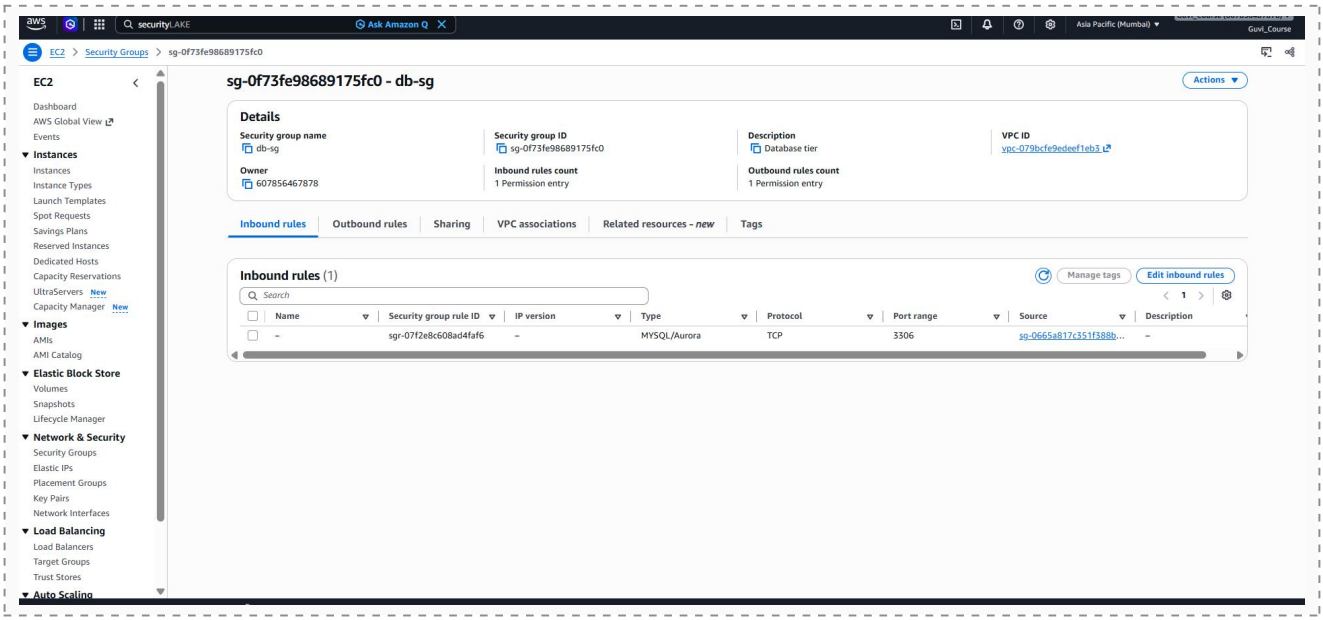





sg-05b97e2bcc1315b3c - web-sg
Actions
Details
Security group name
web-sg
Security group ID
sg-05b97e2bcc1315b3c
Description
Web tier
VPC ID
vpc-079bcfe9edeef1eb3
Owner
607856467878
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
Description
sgr-0e6984db58d56f28b
HTTP
TCP
80

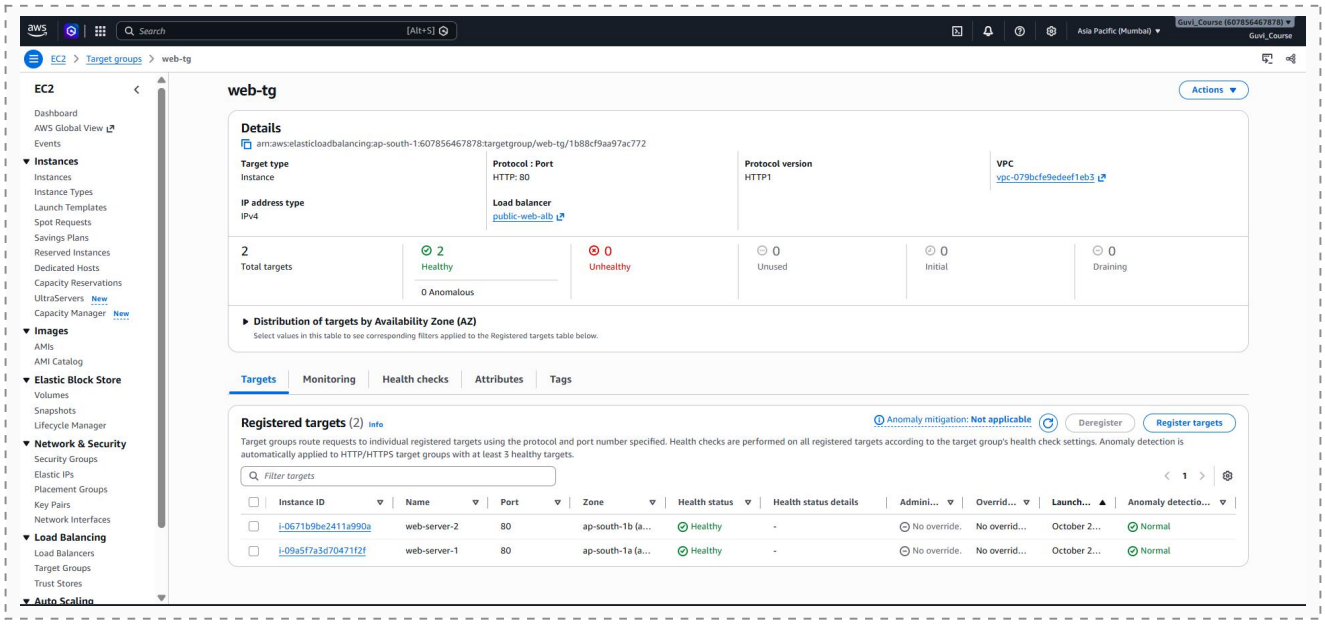

sg-055c26d79a90146e7 - internal-alb-sg
Actions
Details
Security group name
internal-alb-sg
Security group ID
sg-055c26d79a90146e7
Description
Internal ALB
VPC ID
vpc-079bcfe9edeef1eb3
Owner
607856467878
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
Description
sgr-09e666c3fca654034
HTTP
TCP
80

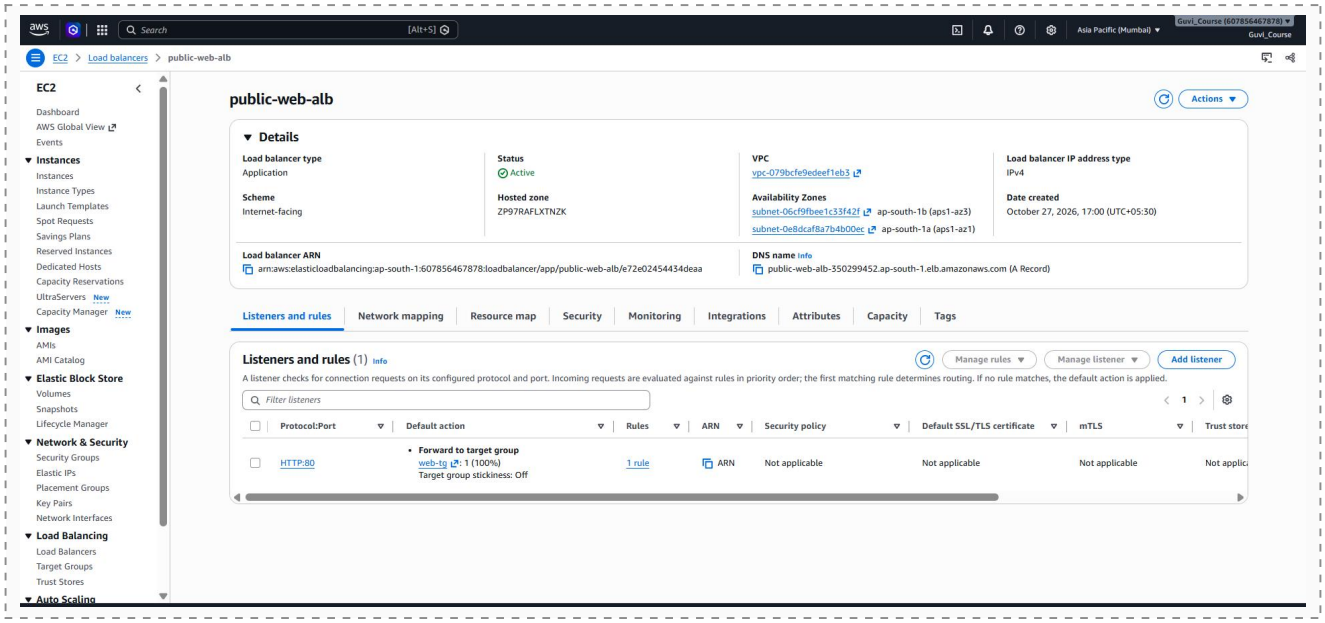

sg-0665a817c351f388b - app-sg
Actions
Details
Security group name
app-sg
Security group ID
sg-0665a817c351f388b
Description
App tier
VPC ID
vpc-079bcfe9edeef1eb3
Owner
607856467878
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
Description
sgr-0098905b762024b96
Custom TCP
TCP
5000


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EC2 > Security Groups > sg-0f73fe98689175fc0
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Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
sg-0f73fe98689175fc0 - db-sg
Actions
Details
Security group name
db-sg
Security group ID
sg-0f73fe98689175fc0
Description
Database tier
VPC ID
vpc-079bcfe9edeef1eb3
Owner
607856467878
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
Description
-
sgr-07f2e8c608ad4faf6
-
MYSQL/Aurora
TCP
3306
sg-0665a817c351f388b...
-

7. Task 6: Internet-facing Application Load Balancer

1. Created the target group web-tg (instances, HTTP 80, health check path /) and registered web-server-1 and web-server-2.
2. Created the internet-facing ALB public-web-alb in public-web-subnet-1a and public-web-subnet-1b with the public-alb-sg security group.
3. Added an HTTP:80 listener forwarding to web-tg.
4. Waited until both targets were Healthy and opened the ALB DNS name in a browser.

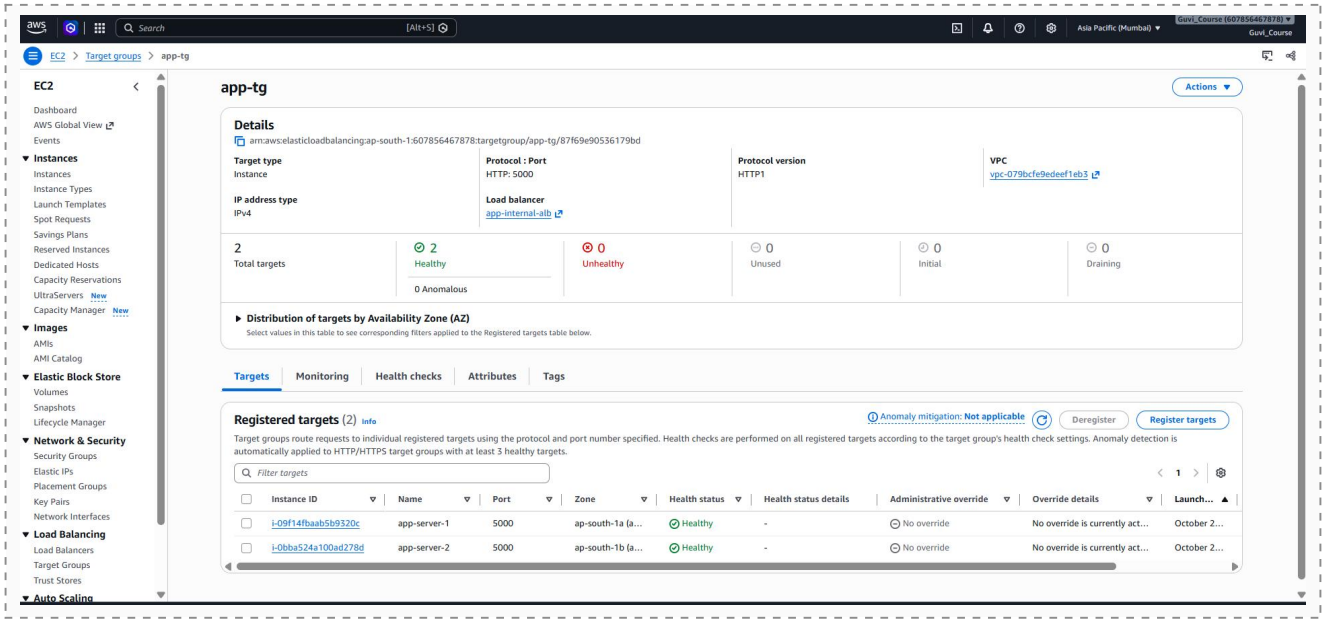


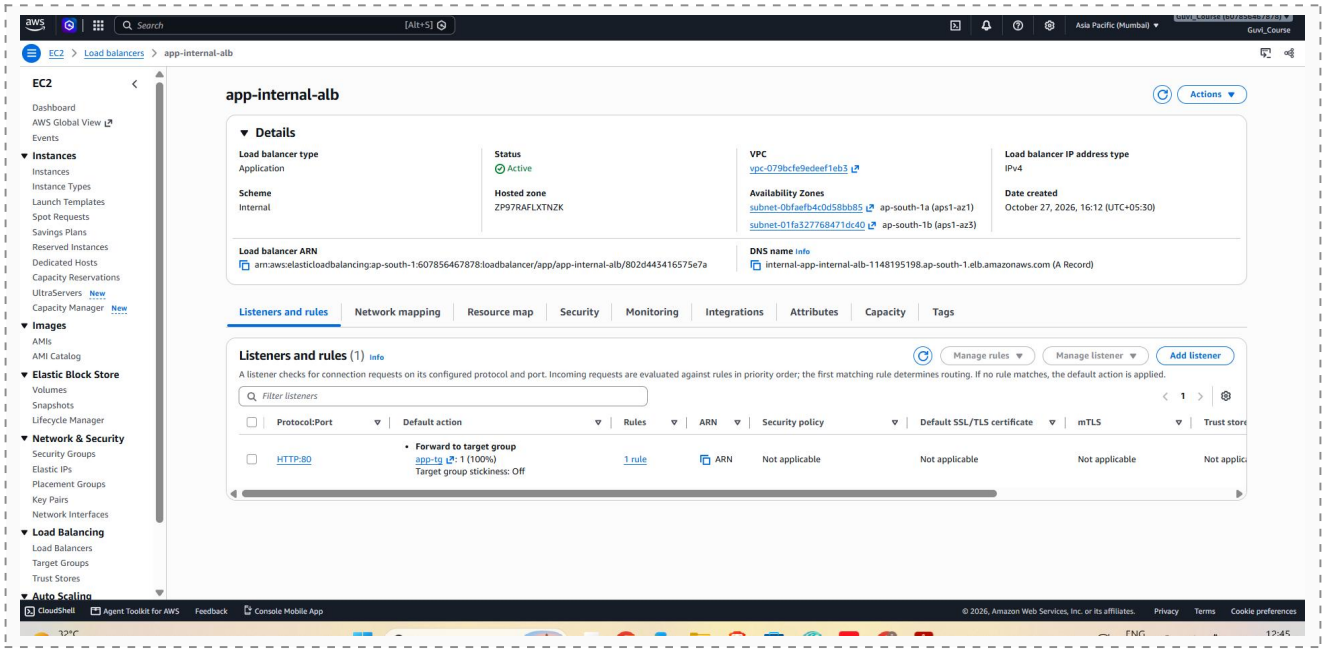


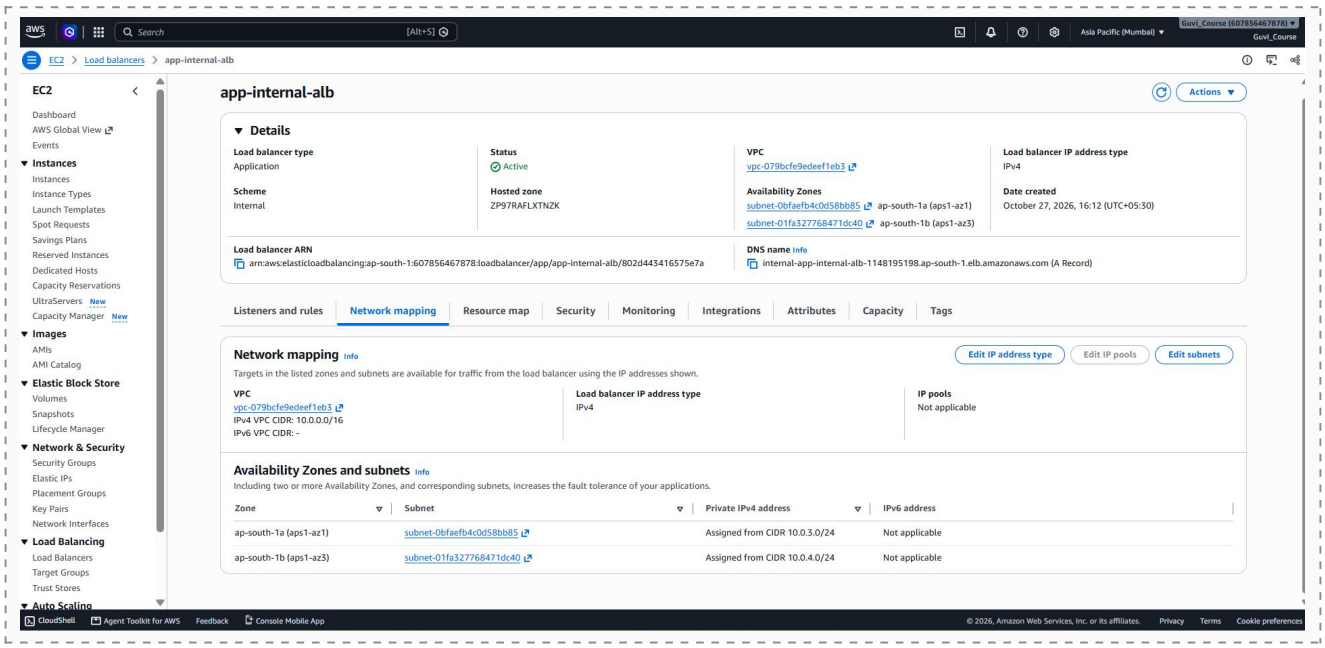
8. Task 7: Internal Application Load Balancer

1. Created the target group app-tg (instances, HTTP 5000, health check path /) and registered app-server-1 and app-server-2.
2. Created the internal ALB app-internal-alb in private-app-subnet-1a and private-app-subnet-1b with the internal-alb-sg security group.
3. Added an HTTP:80 listener forwarding to app-tg.
4. Confirmed both Flask targets were Healthy.

Internal ALB DNS name: internal-app-internal-alb-1148195198.ap-south-1.elb.amazonaws.com






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Guvi_Course
EC2 > Load balancers > app-internal-alb
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Instances
Instance Types
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Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
app-internal-alb
Actions
Details
Load balancer type
Application
Status
Active
VPC
vpc-079bcfe9edeef1eb3
Load balancer IP address type
IPv4
Scheme
Internal
Hosted zone
ZP97RAFLXTNZK
Availability Zones
subnet-0bfaefb4c0d58bb85 ap-south-1a (aps1-az1)
subnet-01fa327768471dc40 ap-south-1b (aps1-az3)
Date created
October 27, 2026, 16:12 (UTC+05:30)
Load balancer ARN
arn:aws:elasticloadbalancing:ap-south-1:607856467878:loadbalancer/app/app-internal-alb/802d443416575e7a
DNS name Info
internal-app-internal-alb-1148195198.ap-south-1.elb.amazonaws.com (A Record)
Listeners and rules
Network mapping
Resource map
Security
Monitoring
Integrations
Attributes
Capacity
Tags
Network mapping Info
Edit IP address type
Edit IP pools
Edit subnets
Targets in the listed zones and subnets are available for traffic from the load balancer using the IP addresses shown.
VPC
vpc-079bcfe9edeef1eb3
IPv4 VPC CIDR: 10.0.0.0/16
IPv6 VPC CIDR: -
Load balancer IP address type
IPv4
IP pools
Not applicable
Availability Zones and subnets Info
Including two or more Availability Zones, and corresponding subnets, increases the fault tolerance of your applications.
Zone
Subnet
Private IPv4 address
IPv6 address
ap-south-1a (aps1-az1)
subnet-0bfaefb4c0d58bb85
Assigned from CIDR 10.0.3.0/24
Not applicable
ap-south-1b (aps1-az3)
subnet-01fa327768471dc40
Assigned from CIDR 10.0.4.0/24
Not applicable
CloudShell
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Feedback
Console Mobile App
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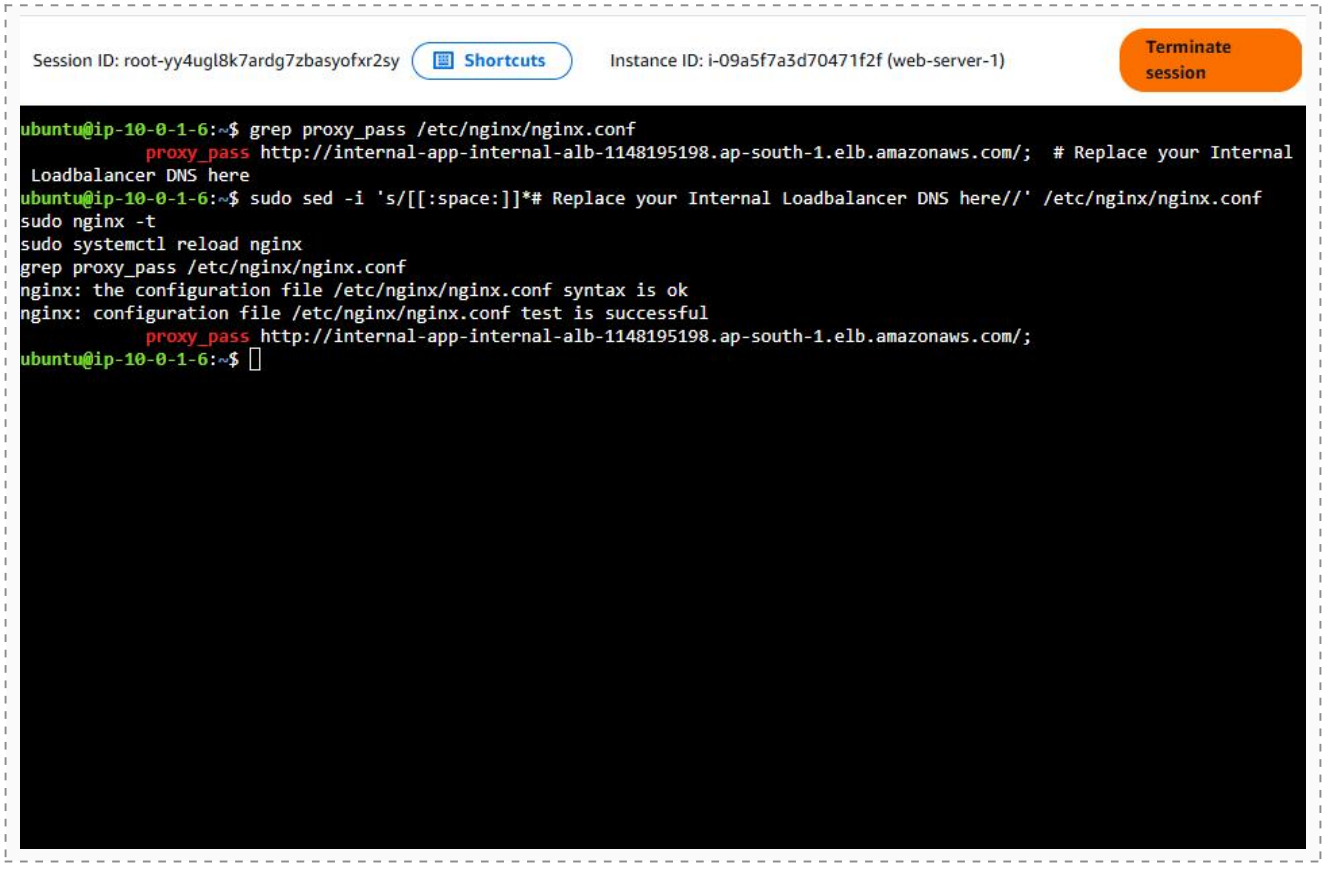
9. Task 8: Nginx reverse proxy for API routing

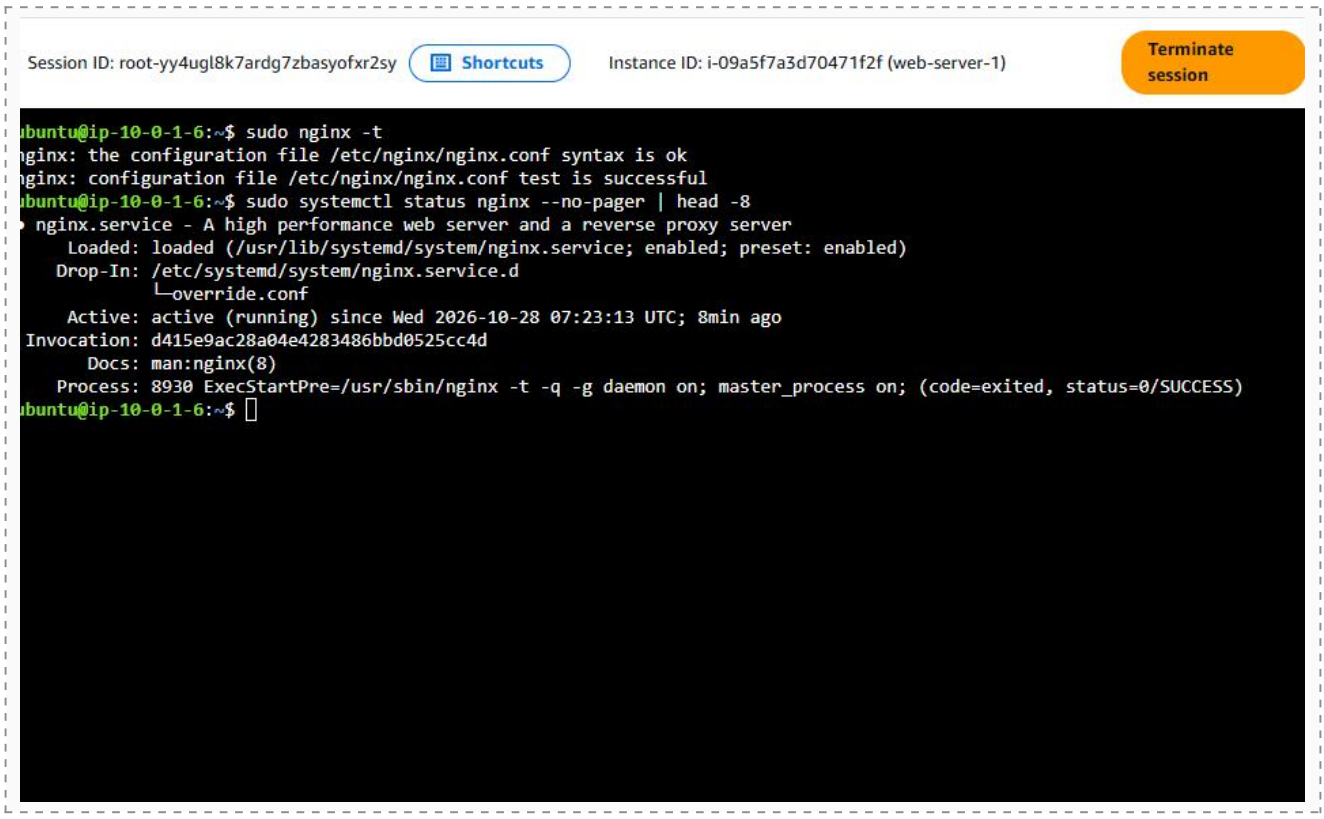
The provided nginx.conf serves the form from /usr/share/nginx/html and forwards requests under /api/ to the internal ALB. The placeholder DNS name was replaced with the real internal ALB DNS name. The trailing slash in proxy_pass makes Nginx strip the /api/ prefix, so /api/submit reaches Flask as /submit.

Commands run on web-server-1

```
sed -i 's/internal-private-Loadbalancer-DNS/internal-app-internal-alb-1148195198.ap-south-1.elb.amazonaws.com/' Three-Tier-Architecture/nginx.conf
grep proxy_pass Three-Tier-Architecture/nginx.conf
sudo cp Three-Tier-Architecture/nginx.conf /etc/nginx/nginx.conf
sudo nginx -t
sudo systemctl restart nginx
curl localhost/api/
```

Result: nginx -t reported the syntax is ok and the test is successful, Nginx restarted as active (running), and curl localhost/api/ returned {"status":"Flask is running"}, which proves the web tier reaches the application tier through the internal ALB.




Session ID: root-yy4ugl8k7ardg7zbasyofxr2sy
Shortcuts
Instance ID: i-09a5f7a3d70471f2f (web-server-1)
Terminate session
ubuntu@ip-10-0-1-6:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@ip-10-0-1-6:~$ sudo systemctl status nginx --no-pager | head -8
nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
Drop-In: /etc/systemd/system/nginx.service.d
override.conf
Active: active (running) since Wed 2026-10-28 07:23:13 UTC; 8min ago
Invocation: d415e9ac28a04e4283486bbd0525cc4d
Docs: man:nginx(8)
Process: 8930 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
ubuntu@ip-10-0-1-6:~$


Session ID: root-yy4ugl8k7ardg7zbasyofxr2sy
Shortcuts
Instance ID: i-09a5f7a3d70471f2f (web-server-1)
Terminate session
ubuntu@ip-10-0-1-6:~$ curl localhost/api/
{"status":"Flask is running"}
ubuntu@ip-10-0-1-6:~$

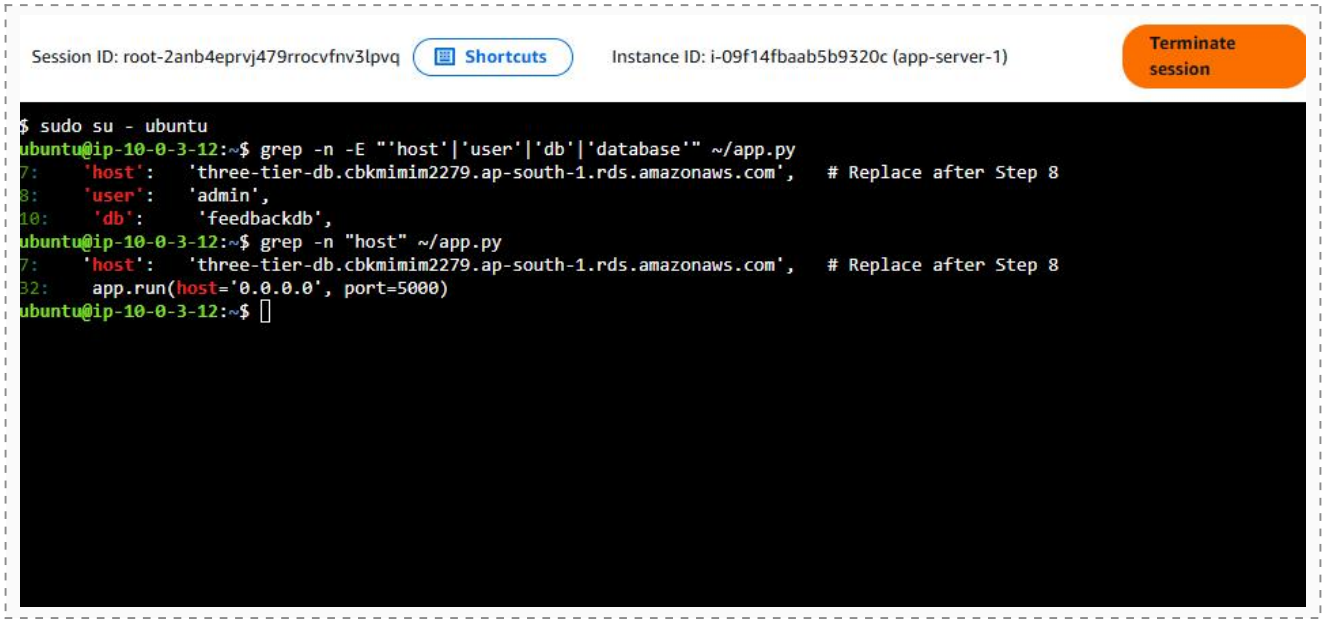
10. Task 9: Flask and Amazon RDS integration

The RDS-ENDPOINT placeholder in app.py was replaced with the RDS endpoint on both app servers. MySQL 8.4 uses caching_sha2_password, so the cryptography package was installed into the virtual environment for PyMySQL. The service was then restarted.

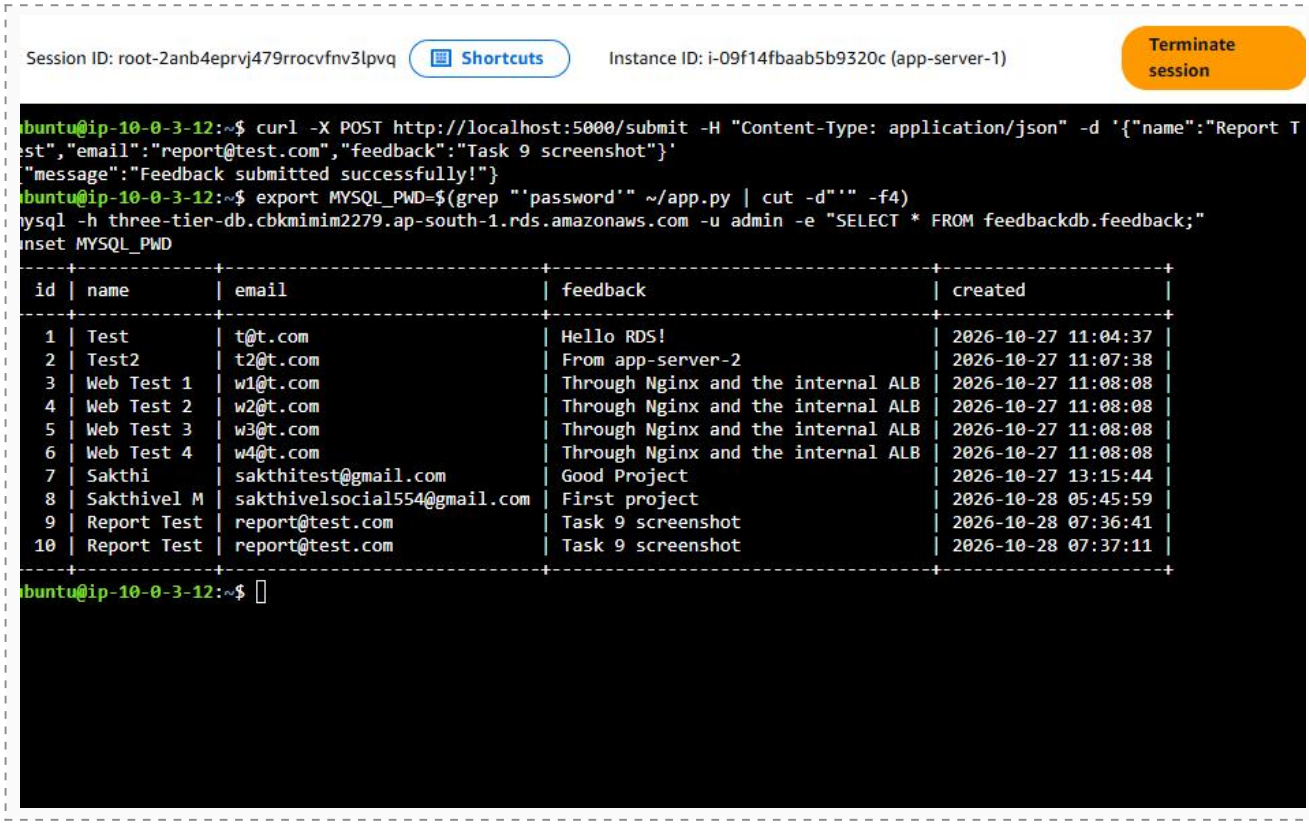
```
sed -i "s/RDS-ENDPOINT/three-tier-db.cbkmimim2279.ap-south-1.rds.amazonaws.com/" ~/app.py
source ~/venv/bin/activate
pip install cryptography
sudo systemctl restart flask-app

curl -X POST http://localhost:5000/submit -H "Content-Type: application/json" \
  -d '{"name":"Test","email":"t@t.com","feedback":"Hello RDS!"}'
# {"message":"Feedback submitted successfully!"}
```

Test submissions were also sent through Nginx and the internal ALB (curl http://localhost/api/submit on web-server-1) so they were distributed across both app servers, and all returned the success message.

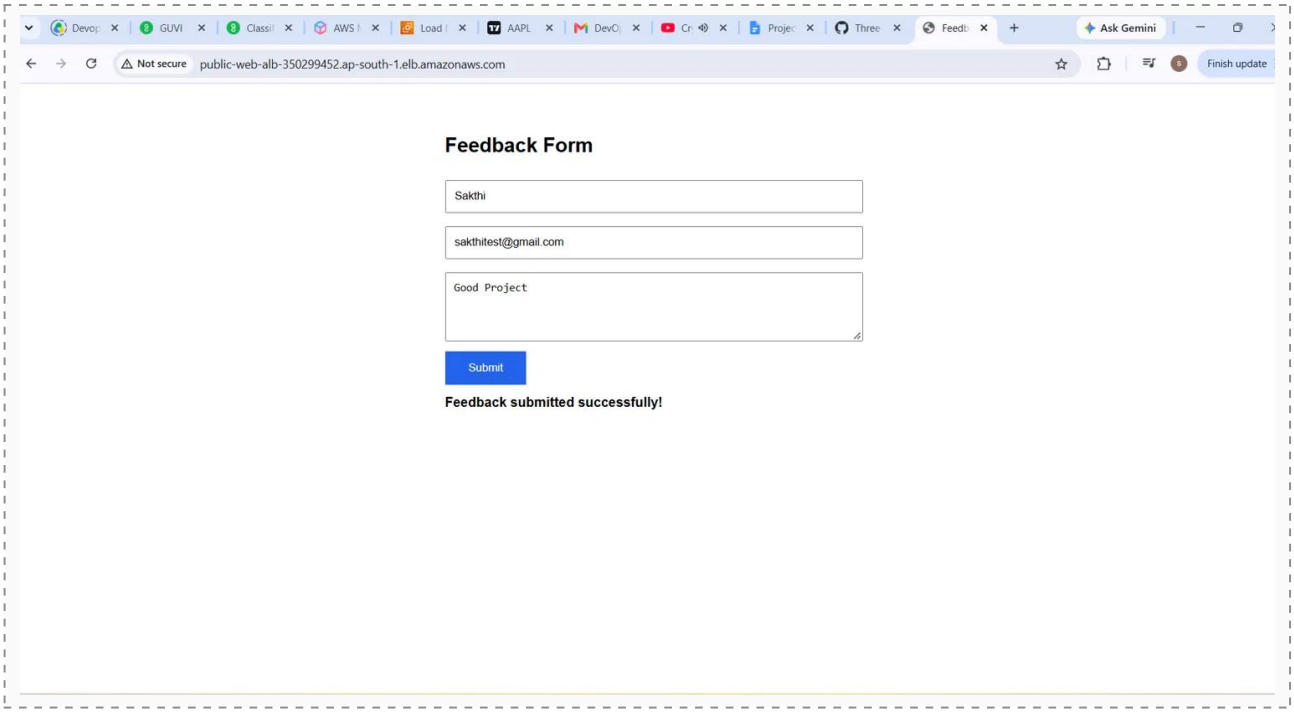


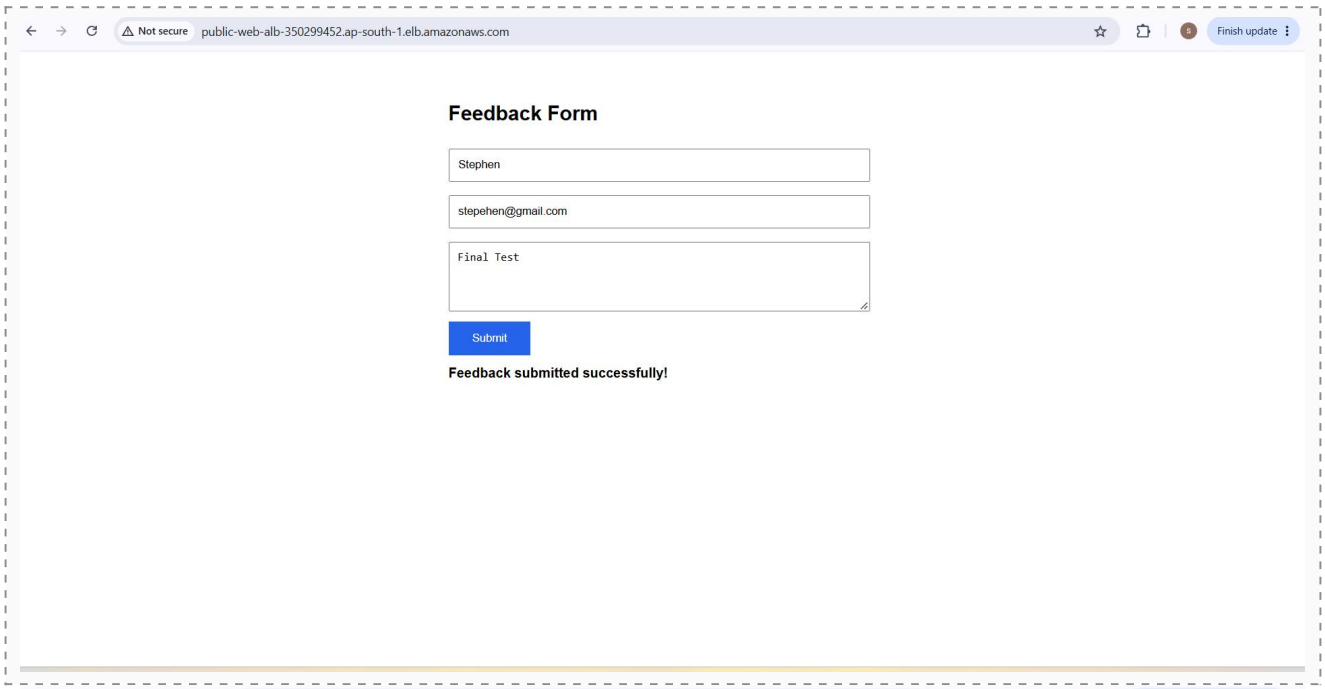


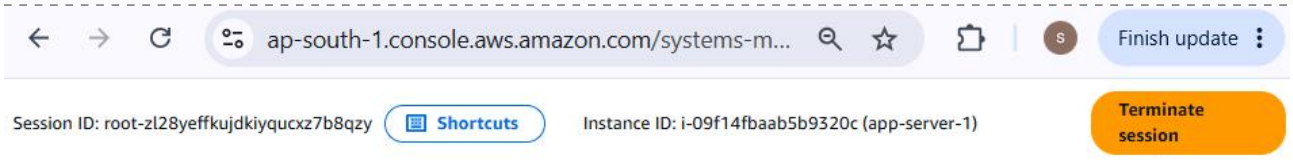

Session ID: root-2anb4eprvj479rrocvfnv3lpvq
Shortcuts
Instance ID: i-09f14fbaab5b9320c (app-server-1)
Terminate session
buntu@ip-10-0-3-12:~$ curl -X POST http://localhost:5000/submit -H "Content-Type: application/json" -d '{"name":"Report T
st","email":"report@test.com","feedback":"Task 9 screenshot"}'
"message":"Feedback submitted successfully!"}
buntu@ip-10-0-3-12:~$ export MYSQL_PWD=$(grep "'password'" ~/app.py | cut -d"'" -f4)
ysql -h three-tier-db.cbkmimim2279.ap-south-1.rds.amazonaws.com -u admin -e "SELECT * FROM feedbackdb.feedback;"
nset MYSQL_PWD
id | name | email | feedback | created
1 | Test | t@t.com | Hello RDS! | 2026-10-27 11:04:37
2 | Test2 | t2@t.com | From app-server-2 | 2026-10-27 11:07:38
3 | Web Test 1 | w1@t.com | Through Nginx and the internal ALB | 2026-10-27 11:08:08
4 | Web Test 2 | w2@t.com | Through Nginx and the internal ALB | 2026-10-27 11:08:08
5 | Web Test 3 | w3@t.com | Through Nginx and the internal ALB | 2026-10-27 11:08:08
6 | Web Test 4 | w4@t.com | Through Nginx and the internal ALB | 2026-10-27 11:08:08
7 | Sakthi | sakthitest@gmail.com | Good Project | 2026-10-27 13:15:44
8 | Sakthivel M | sakthivelsocial554@gmail.com | First project | 2026-10-28 05:45:59
9 | Report Test | report@test.com | Task 9 screenshot | 2026-10-28 07:36:41
10 | Report Test | report@test.com | Task 9 screenshot | 2026-10-28 07:37:11
buntu@ip-10-0-3-12:~$

11. Task 10: Validation of the complete architecture

Requirement	Result	Evidence
Web app reachable through the public ALB	Passed	Browser screenshot with the ALB DNS URL
Feedback form accepts input	Passed	Form filled and submitted in the browser
Nginx forwards API requests to the internal ALB	Passed	curl localhost/api/ and /api/submit succeeded
Flask processes requests	Passed	Success message returned; service active
Data stored in RDS MySQL	Passed	Row visible in the feedback table
Data verified with SQL	Passed	SELECT * FROM feedbackdb.feedback;
Only required traffic between tiers	Passed	Security group chain: ALB → web → internal ALB → app → db








```
$ sudo su - ubuntu
ubuntu@ip-10-0-3-12:~$ export MYSQL_PWD=$(grep "'password'" ~/app.py | cut -d"'" -f4)
mysql -h three-tier-db.cbkmimim2279.ap-south-1.rds.amazonaws.com -u admin -e "SELECT * FROM feedbackdb.feedback;"
unset MYSQL_PWD
+----+-------------+-----------------------------+-----------------------------------+---------------------+
| id | name        | email                       | feedback                          | created             |
+----+-------------+-----------------------------+-----------------------------------+---------------------+
|  1 | Test        | t@t.com                     | Hello RDS!                        | 2026-10-27 11:04:37 |
|  2 | Test2       | t2@t.com                    | From app-server-2                 | 2026-10-27 11:07:38 |
|  3 | Web Test 1  | w1@t.com                    | Through Nginx and the internal ALB | 2026-10-27 11:08:08 |
|  4 | Web Test 2  | w2@t.com                    | Through Nginx and the internal ALB | 2026-10-27 11:08:08 |
|  5 | Web Test 3  | w3@t.com                    | Through Nginx and the internal ALB | 2026-10-27 11:08:08 |
|  6 | Web Test 4  | w4@t.com                    | Through Nginx and the internal ALB | 2026-10-27 11:08:08 |
|  7 | Sakthi      | sakthitest@gmail.com        | Good Project                      | 2026-10-27 13:15:44 |
|  8 | Sakthivel M | sakthivelsocial554@gmail.com | First project                     | 2026-10-28 05:45:59 |
|  9 | Report Test | report@test.com             | Task 9 screenshot                 | 2026-10-28 07:36:41 |
| 10 | Report Test | report@test.com             | Task 9 screenshot                 | 2026-10-28 07:37:11 |
| 11 | Stephen     | stepehen@gmail.com          | Final Test                        | 2026-10-28 10:21:37 |
+----+-------------+-----------------------------+-----------------------------------+---------------------+
ubuntu@ip-10-0-3-12:~$
```